AOS-CX 10.17.xxxx Link Aggregation Guide (All Switch Series)

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About this document

This document describes features of the AOS-CX network operating system. It is intended for administrators responsible for installing, configuring, and managing HPE Aruba Networking switches on a network.

Subtopics

Applicable products

This document applies to the following products:

- HPE Aruba Networking 4100i Switch Series (JL817A, JL818A)
- HPE Aruba Networking 5420 Switch Series (S0U67A, S0U55A, S0U63A, S0U64A, S0U65A, S0U75A, S0U72A, S0U78A, S0U58A, S0U73A, S0U74A, S0U71A, S0U76A, S0U70A, S0U77A, S0U60A, S0U61A, S0U62A, S0U66A, S0U68A)
- HPE Aruba Networking 6000 Switch Series (R8N85A, R8N86A, R8N87A, R8N88A, R8N89A, R9Y03A, S4R20A, S4R21A, S4R22A, S4R23A, S4R24A, S4R25A, S4R26A, S4R27A, S4R28, S4R29A)
- HPE Aruba Networking 6100 Switch Series (JL675A, JL676A, JL677A, JL678A, JL679A)
- HPE Aruba Networking 6200 Switch Series (JL724A, JL725A, JL726A, JL727A, JL728A, R8Q67A, R8Q68A, R8Q69A, R8Q70A, R8Q71A, R8V08A, R8V09A, R8V10A, R8V11A, R8V12A, R8Q72A, JL724B, JL725B, JL726B, JL727B, JL728B, S0M81A, S0M82A, S0M83A, S0M84A, S0M85A, S0M86A, S0M87A, S0M88A, S0M89A, S0M90A, S0G13A, S0G14A, S0G15A, S0G16A, S0G17A)
- HPE Aruba Networking 6300 Switch Series (JL658A, JL659A, JL660A, JL661A, JL662A, JL663A, JL664A, JL665A, JL666A, JL667A, JL668A, JL762A, R8S89A, R8S90A, R8S91A, R8S92A, S0E91A, S0X44A, S3L75A, S3L76A, S3L77A, S4P41A,S4P42A, S4P43A, S4P44A, S4P45A, S4P46A, S4P47A, S4P48A)
- HPE Aruba Networking 6400 Switch Series (R0X31A, R0X38B, R0X38C, R0X39B, R0X39C, R0X40B, R0X40C, R0X41A, R0X41C, R0X42A, R0X42C, R0X43A, R0X43C, R0X44A, R0X44C, R0X45A, R0X45C, R0X26A, R0X27A, JL741A, S0E48A,S0E48A #0D1, S1T83A, S1T83A #0D1)

- HPE Aruba Networking 8100 Switch Series (R9W94A, R9W95A, R9W96A, R9W97A)
- HPE Aruba Networking 8320 Switch Series (JL479A, JL579A, JL581A)
- HPE Aruba Networking 8325 Switch Series (JL624A, JL625A, JL626A, JL627A)
- HPE Aruba Networking 8325H Switch Series (S4B20A, S4B21A, S4B22A, S4B23A, S2T42A, S2T46A, S2T47A, S2T48A)
- HPE Aruba Networking 8325P Switch Series (S0G12A, S4A48A)
- HPE Aruba Networking 8360 Switch Series (JL700A, JL701A, JL702A, JL703A, JL706A, JL707A, JL708A, JL709A, JL710A, JL711A, JL700C, JL701C, JL702C, JL703C, JL706C, JL707C, JL708C, JL709C, JL710C, JL711C, JL704C, JL705C, JL719C, JL718C, JL717C, JL720C, JL722C, JL721C)
- HPE Aruba Networking 8400 Switch Series (JL366A, JL363A, JL687A)
- HPE Aruba Networking 9300 Switch Series (R9A29A, R9A30A, R8Z96A, S0F81A, S0F82A, S0F83A)
- HPE Aruba Networking 9300S Switch Series (S0F81A, S0F82A, S0F83A, S0F84A, S0F85A, S0F86A, S0F88A, S0F95A, S0F96A)
- HPE Aruba Networking 10000 Switch Series (R8P13A, R8P14A)
- HPE Aruba Networking 10040 Switch Series (S4R58A, S4R59A)

Latest version available online

Updates to this document can occur after initial publication. For the latest versions of product documentation, see the links provided in Support and Other Resources.

Command syntax notation conventions

Convention	Usage
example - text	Identifies commands and their options and operands , code examples, filenames, pathnames, and output displayed in a command window. Items that appear like the example text in the previous column are to be entered exactly as shown and are required unless enclosed in brackets (**[ ]**).

<table>
<tr><th>Convention</th><th>Usage</th></tr>
<tr><td>example - text</td><td>In code and screen examples, indicates text entered by a user.</td></tr>
<tr><td>Any of the following:
• <example - text>
• <example - text>
• example - text
• example - text</td><td>Identifies a placeholder—such as a parameter or a variable—that you must substitute with an actual value in a command or in code:
• For output formats where italic text cannot be displayed, variables are enclosed in angle brackets (< >). Substitute the text—including the enclosing angle brackets—with an actual value.
• For output formats where italic text can be displayed, variables might or might not be enclosed in angle brackets. Substitute the text including the enclosing angle brackets, if any, with an actual value.</td></tr>
<tr><td>|</td><td>Vertical bar. A logical OR that separates multiple items from which you can choose only one.
Any spaces that are on either side of the vertical bar are included for readability and are not a required part of the command syntax.</td></tr>
<tr><td>{ }</td><td>Braces. Indicates that at least one of the enclosed items is required.</td></tr>
<tr><td>[ ]</td><td>Brackets. Indicates that the enclosed item or items are optional.</td></tr>
<tr><td>… or
...</td><td>Ellipsis:
• In code and screen examples, a vertical or horizontal ellipsis indicates an omission of information.
• In syntax using brackets and braces, an ellipsis indicates items that can be repeated. When an item followed by ellipses is enclosed in brackets, zero or more items can be specified.</td></tr>
</table>

About the examples

Examples in this document are representative and might not match your particular switch or environment.

The slot and port numbers in this document are for illustration only and might be unavailable on your switch.

Understanding the CLI prompts

When illustrating the prompts in the command line interface (CLI), this document uses the generic term **switch**, instead of the host name of the switch. For example:

```
switch>
```

The CLI prompt indicates the current command context. For example:

```
switch>
```

Indicates the operator command context.

```
switch#
```

Indicates the manager command context.

```
switch(CONTEXT-NAME)#
```

Indicates the configuration context for a feature. For example:

```
switch(config-if)#
```

Identifies the **interface** context.

Variable information in CLI prompts

In certain configuration contexts, the prompt may include variable information. For example, when in the VLAN configuration context, a VLAN number appears in the prompt:

```
switch(config-vlan-100)#
```

When referring to this context, this document uses the syntax:

```
switch(config-vlan-<VLAN-ID>)#
```

Where <VLAN-ID> is a variable representing the VLAN number.

Identifying switch ports and interfaces

Physical ports on the switch and their corresponding logical software interfaces are identified using the format: **member/slot/port.**

On the HPE Aruba Networking 4100i Switch Series

- member: Always 1. VSF is not supported on this switch.
- slot: Always 1. This is not a modular switch, so there are no slots.
- port: Physical number of a port on the switch.

For example, the logical interface **1/1/4** in software is associated with physical port 4 on the switch.

On the HPE Aruba Networking 6000 and 6100 Switch Series

- member: Always 1. VSF is not supported on this switch.
- slot: Always 1. This is not a modular switch, so there are no slots.
- port: Physical number of a port on the switch.

For example, the logical interface **1/1/4** in software is associated with physical port 4 on the switch.

On the HPE Aruba Networking 6200 Switch Series

- member: Member number of the switch in a Virtual Switching Framework (VSF) stack. Range: 1 to 8. The primary switch is always member 1. If the switch is not a member of a VSF stack, then member is 1.
- slot: Always 1. This is not a modular switch, so there are no slots.
- port: Physical number of a port on the switch.

For example, the logical interface **1/1/4** in software is associated with physical port 4 in slot 1 on member 1.

On the HPE Aruba Networking 6300 Switch Series

- member: Member number of the switch in a Virtual Switching Framework (VSF) stack. Range: 1 to 10. The primary switch is always member 1. If the switch is not a member of a VSF stack, then member is 1.
- slot: Always 1. This is not a modular switch, so there are no slots.
- port: Physical number of a port on the switch.

For example, the logical interface **1/1/4** in software is associated with physical port 4 on member 1.

On the HPE Aruba Networking 6400 and 5420 Switch Series

- member: Always 1. VSF is not supported on this switch.
- slot: Specifies physical location of a module in the switch chassis.
  - Management modules are on the front of the switch in slots 1/1 and 1/2.
  - Line modules are on the front of the switch starting in slot 1/3.
- port: Physical number of a port on a line module.

For example, the logical interface **1/3/4** in software is associated with physical port 4 in slot 3 on member 1.

On the HPE Aruba Networking 8xxx, 93xx, and 10xxx Switch Series

- member: Always 1. VSF is not supported on this switch.
- slot: Always 1. This is not a modular switch, so there are no slots.
- port: Physical number of a port on the switch.

For example, the logical interface **1/1/4** in software is associated with physical port 4 on the switch.

> **NOTE**
> If using breakout cables, the port designation changes to x:y, where x is the physical port and y is the lane when split. For example, the logical interface 1/1/4:2 in software is associated with lane 2 on physical port 4 in slot 1 on member 1.

On the HPE Aruba Networking 8400 Switch Series

- member: Always 1. VSF is not supported on this switch.
- slot: Specifies physical location of a module in the switch chassis.
    - Management modules are on the front of the switch in slots 1/5 and 1/6.
    - Line modules are on the front of the switch in slots 1/1 through 1/4, and 1/7 through 1/10.
- port: Physical number of a port on a line module

For example, the logical interface **1/1/4** in software is associated with physical port 4 in slot 1 on member 1.

Identifying modular switch components

- Power supplies are on the front of the switch behind the bezel above the management modules. Power supplies are labeled in software in the format: member/power supply:
    - member: 1.
    - power supply: 1 to 4.
- Fans are on the rear of the switch and are labeled in software as: member/tray/fan:
    - member: 1.
    - tray: 1 to 4.
    - fan: 1 to 4.
- Fabric modules are not labeled on the switch but are labeled in software in the format: member/module:
    - member: 1.
    - member: 1 or 2.
- The display module on the rear of the switch is not labeled with a member or slot number.

Link Aggregation

Subtopics

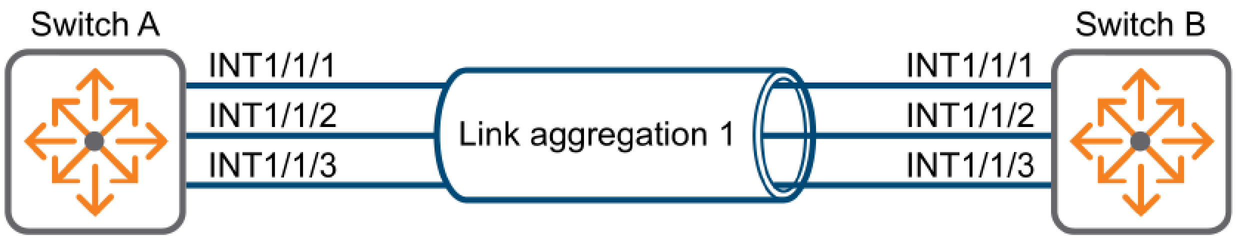
Overview

Ethernet link aggregation bundles multiple physical Ethernet links into one logical link, called a link aggregation group (LAG).

Link aggregation has the following benefits:

- Increased bandwidth beyond the limits of any single link. In an aggregate link, traffic is distributed across the member ports.
- Improved link reliability. The member ports dynamically back up one another. When a member port fails, its traffic is automatically switched to other member ports. As shown in the following figure Device A and Device B are connected by three physical Ethernet links. These physical Ethernet links are combined into an aggregate link called link aggregation 1. The bandwidth of this aggregate link can reach up to the total bandwidth of the three physical Ethernet links. At the same time, the three Ethernet links back up one another. When a physical Ethernet link fails, the traffic originally intended for the failed link is switched to the remaining active links.

Ethernet link aggregation diagram



Switch Capacities Per Platform

Platform	Max interfaces per LAG interface	Max interfaces per multi - chassis LAG interface	Max LAG interfac e configurations all owed	Max multi - chassis LAG interfaces confi guration allowed
10000	16	16	128	128
9300/9300S	16	16	128	128
8400	16	16	256	256
8320	16	16	128	128
8325/8325H	16	16	54	54
8325P	16	16	128	128
5420	16	16	144	144
6000/6100	8	0	8	0
4100i	8	0	8	0
6200/6200v2	16	16	48	32
6300M/F	16	16	256	32
6300L	16	0	256	0
6400	16	16	480	480

Aggregation group, member port, and aggregate interface

An aggregation group is a collection of physical interfaces that are bundled together for the purpose of load distribution and redundancy. These physical interfaces are called member ports. They are configured through a logical aggregate interface.

An aggregate interface can be one of the following types:

- **Layer 2**: The member ports of the corresponding Link Aggregation Group can only be layer 2 Ethernet interfaces.
- **Layer 3**: The member ports of the corresponding Link Aggregation Group can only be layer 3 interfaces.

> **NOTE**
>
> Layer 3 aggregation groups are not supported on the 4100i, 5420, 6000, 6100, and 6200 Switch Series.

The effective port rate of an aggregate interface equals the total rate of its member ports. Only full duplex mode members are eligible for aggregation.

Link aggregation modes

An aggregation group operates in one of the following modes:

- **Static LAG**: In the static LAG mode of operation, Link failure is not detected as there is no keep alive PDU communication between the devices. A misconfiguration on one side can cause much trouble and be difficult to troubleshoot, because no signaling takes place between the two peers.
- **Dynamic LAG or LACP**: The local device and the peer device automatically maintain the aggregation states of the member ports, resulting in link failure being quickly detected by exchanging the PDU. LACP reduces the workload of network administrators.
  Dynamic LAG uses LACP packets to establish the association between two peers. This configuration results in the reduction of the misconfiguration probability. Also, link failures are intelligently handled by two participating devices through the LACP protocol, which is adaptive or dynamic to these network failures.

  Layer 2 aggregation groups and layer 3 aggregation groups support both static and dynamic modes.

LACP

Dynamic aggregation is implemented through the IEEE 802.3ad Link Aggregation Control Protocol (LACP).

LACP uses LACPDUs to exchange aggregation information between LACP-enabled devices. Each member port in a dynamic aggregation group can exchange information with its peer. When a member port receives an LACPDU, it compares the received information with information received on the other member ports. In this way, the two systems agree on which ports are placed in Selected state.

The LACPDU fields convey data for the LACP functions, including:

- System LACP priority
- System MAC address
- Port priority
- Port number
- Operational key

Subtopics

LACP operating modes

LACP can operate in active or passive mode.

- **Active mode:** When the LACP is operating in active mode on either end of a link, both ports can send PDUs. The "active" LACP initiates an LACP connection by sending LACPDUs. The "passive" LACP will wait for the remote end to initiate the link.
- **Passive mode:** When the LACP is operating in passive mode on a local member port and as its peer port, both ports cannot send PDUs.

> **NOTE**
>
> Two peer ports operating in "passive" mode will never establish an LACP link.

For an LACP LAG, one side must have LACP in active mode and the peer must have an LACP configuration of active or passive mode. If you do not enable LACP on a LAG, it is treated as a static LAG and the peer cannot negotiate LACP with the LAG.

LACP configuration settings

Task	Command	Example
Setting the LACP mode to active or passive.	`lacp mode {active | passive}`	`switch(config-lag-if)# lacp mode active`
Setting the LACP mode to off.	`no lacp mode {active | passive}`	`switch(config-lag-if)# no lacp mode active`

Setting the hash type.	**For 6000, 6100, and 8400 Switch Series:**
	`lacp hash [l2-src-dst | l3-src-dst | l4-src-dst]`
	For 8320, 8325/8325P/8325H, 5420, 6200, 6300, 6400 and 10000 Switch Series:
	`hash [l2-src-dst | l3-src-dst | l4-src-dst]`
		For 6000, 6100, and 8400 Switch Series:
		`switch(config)# lacp hash l2-src-dst`
		For 8320, 8325/8325P/8325H, 5420, 6200, 6300, 6400 and 10000 Switch Series:
		`switch(config-lag-if)# hash l2-src-dst`

Setting the LACP rate to fast.	`lacp rate fast`	`switch(config)# interface lag 1` `switch(config-lag-if)# lacp rate fast`
Setting the LACP rate to slow.	`lacp rate slow`	`switch(config)# interface lag 1` `switch(config-lag-if)# lacp rate slow`
Applying shutdown on the LAG port.	`shutdown`	`switch(config)# interface lag 1` `switch(config-lag-if)# shutdown`
Resetting every interface in the LAG to the default (up)	`no shutdown`	`switch(config-lag-if)# no shutdown`

Interface LACP settings

Task	Command	Example
Setting the LACP port ID.	`lacp port-id <ID>`	`switch(config-if)# lacp port-id 100`
Setting the LACP port ID to the default.	`no lacp port-id`	`switch(config-if)# no lacp port-id`
Setting the LACP port priority.	`lacp port-priority <PORT-PRIORITY>`	`switch(config-if)# lacp port-priority 100`
Setting the LACP port priority to the default	`no lacp port-priority`	`switch(config-if)# no lacp port-priority`

LAG interface states

The output from the CLI commands `show lacp interfaces` and `show lacp interfaces multi-chassis` display the following interface states:

Interface state	Description
`A - Active`	An active LACP interface.
`C - Collecting`	Data frames are received through the aggregate link and sent onto the intended destination.
`D - Distributing`	Data frames are transmitted through the aggregate link to reach the intended destination.
`F - Aggregable`	The link can be used as part of an aggregate.
`E - Default neighbor state`	The link has the default state of the neighbor switch.
`I - Individual`	The link is used as an individual link.
`L - Long-timeout`	With the long timeout, an LACPDU is sent every 30 seconds. If no response comes from its partner after three LACPDUs are sent (90 seconds), a timeout event occurs. The LACP state machine then transitions to the appropriate state based on its current state.

Interface state	Description
`N - InSync`	The physical port is connected to the aggregate po rt that was last chosen by the logical election. The state variable selected is still true.
`O - OutofSync`	The hardware might be out of sync with the modifi ed protocol information. If the hardware also has a status of collecting, do not transmit frames because t hey will be misdelivered.
`P - Passive`	The port participates in the protocol, as long as it ha s an active partner.
`S - Short-timeout`	In the short timeout configuration, an LACPDU is sen t every second. If no response comes from its partne r after three LACPDUs are sent, a timeout event occu rs. The LACP state machine then transitions to the a ppropriate state based on its current state.
`X - State m/c expired`	The "current while" timer has expired. The "current w hile" timer then restarts with the short - timeout ena bled. The term `State m/c` refers to a state machine.

How static link aggregation groups are built

Reference port selection process

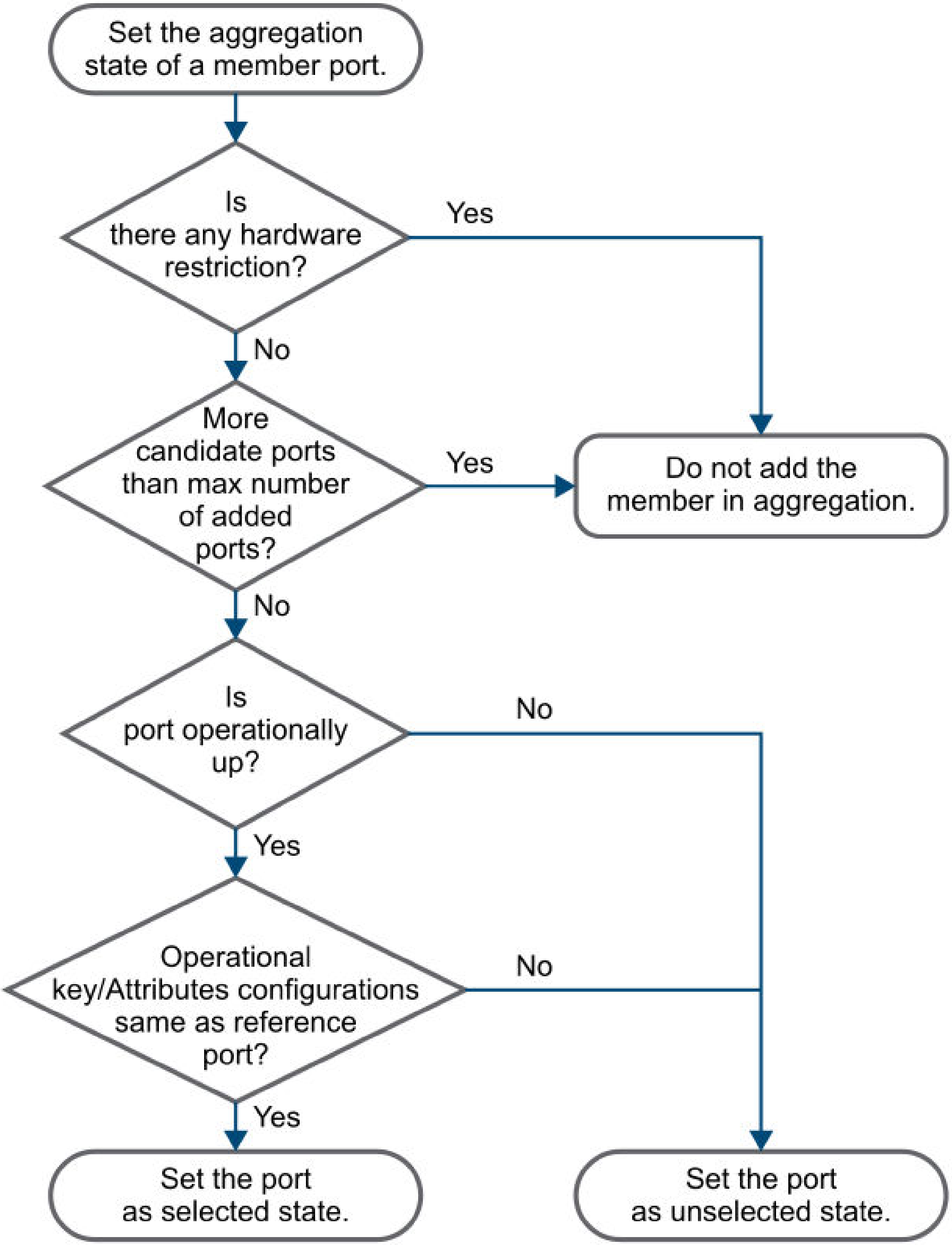
When setting the aggregation states of the ports in an aggregation group, the system automatically chooses a member port as the reference port. A selected port must have the same operational key and attribute configurations as the reference port.

The system chooses a reference port from the member ports in the up state. The first member interface which is operationally up is selected as reference port.

Setting the aggregation state of each member port

After the reference port is chosen, the system sets the aggregation state of each member port in the static aggregation group.

Setting the aggregation state of a member port in a static aggregation group



After the maximum limit of members is reached in a LAG, an additional port cannot be added to the aggregation group. If a port belongs to a card type with a different speed than the other aggregation members, the port can still be added to the aggregation group. If dynamic LAG is enabled, any port member with a speed different than other aggregation members is blocked or ineligible from the same aggregation group. Any operational keys/attributes or configuration changes might affect the aggregation states of the member ports.

How dynamic link aggregation groups are built

Choosing a reference port

About this task

The system chooses a reference port from the member ports in up state. A selected port must have the same operational key and attribute configurations as the reference port.

The process by which the local system (the actor) and the peer system (the partner) negotiate a reference port occurs as follows:

Procedure

1. The two systems determine the system with the smaller system ID. A system ID contains the system LACP priority and the system MAC address.
   a. The two systems compare their LACP priority values.The lower the LACP priority, the smaller the system ID. If the LACP priority values are the same, the two systems proceed to step b.
   b. The two systems compare their MAC addresses.The lower the MAC address, the smaller the system ID.
2. The system with the smaller system ID chooses the first operationally up port as the reference port. A port ID contains a port priority and a port number. The lower the port priority, the smaller the port ID.

Setting the aggregation state of each member port

After the reference port is chosen, the system with the smaller system ID sets the state of each member port on its side.

The system with the greater system ID can detect the aggregation state changes on the peer system. The system with the greater system ID sets the aggregation state of local member ports the same as their peer ports.

When you aggregate interfaces in dynamic mode, follow these guidelines:

- A dynamic link aggregation group chooses only full-duplex ports as the selected ports.
- For stable aggregation and service continuity, do not change the operational key or attribute configurations on any member port.

LAG configuration guidelines

Aggregation member interface restrictions

- If any features in the following list are configured on the interface, you cannot assign an interface to a Layer 2 aggregation group:
    - MAC authentication
    - Port security
    - 802.1X
- Do not assign a reflector port for port mirroring to an aggregation group.

Requirements for adding interfaces

Keep in mind the following requirements when adding interfaces to a LAG:

- To determine the maximum number of LAG interfaces for your type of switch, look at the output from the `show capacities lag` command; however, the number of LAGs that can be created depends on the availability of the physical interface since each LAG interface needs at least one physical interface as a member link. fter the maximum limit of members is reached in a LAG, an additional port cannot be added to the aggregation group. If a port belongs to a card type with a different speed than the other aggregation members, the port can still be added to the aggregation group. If dynamic LAG is enabled, any port member with a speed different than other aggregation members is blocked or ineligible from the same aggregation group. Any operational keys/attributes or configuration changes might affect the aggregation states of the member ports.
- The nondefaults configuration on an interface is removed automatically when the interface is added to a link aggregation. For example: Assume that you remove a member interface from an existing LAG and add it to another LAG. The software removes the nondefault configurations on the interface when it is added to the new LAG.

Configuration consistency requirements

- Configure at least one active mode aggregation in two devices.
- For a successful static aggregation, make sure the ports at both ends of each link are in the same aggregation state.
- For a successful dynamic aggregation, make sure the peer ports of the ports aggregated at one end are also aggregated, and that one of the ends is configured as "active". The two ends can automatically negotiate the aggregation state of each member port.

Removing interfaces

- Deleting an aggregate interface also deletes its aggregation group and causes all member ports to leave the aggregation group.
- When a member interface is removed from a LAG:
  - **4100i, 5420, 6000, 6100, 6200, 6300, and 6400 switches:** The interface goes to its default status of `unshut`.
  - **8100, 8320, 8325/8325P/8325H, 8360, 8400, 9300, or 10000 switches:** The interface becomes disabled.

Disabling an interface

When an interface LAG is disabled with the `shutdown` command, all its members also become operationally down.

Layer 2 aggregation groups

All switches support static and dynamic layer 2 aggregation groups.

> **NOTE**
> On the 6400 Switch Series, port identification differs. Line card ports start at `1/3/1`.

Subtopics

[Configuring a layer 2 static aggregation group]
[Configuring a layer 2 dynamic aggregation group]

Configuring a layer 2 static aggregation group

Prerequisites

You must be in the global configuration context: `switch(config)#`.

Procedure

1. Create a layer 2 aggregate interface and access the layer 2 aggregate interface view by entering:
```
switch(config)# interface lag <ID>
```

The range of the LAG interface ID is 1 to 256.

While creating the layer 2 aggregate interface, the system automatically creates a layer 2 static aggregation group numbered the same.

2. Set the operational state of every interface in the LAG to up by entering:

```
switch(config-lag-if)# no shutdown
```

> **NOTE**
>
> This command does not impact the administrative state of the member interfaces because the command was entered at the level of the LAG. To change the administrative state of a member interface, enter the command at the interface level. For example:
>
> ```
> switch(config)# interface 1/1/2
> switch(config-if)# no shutdown
> ```

3. On the 8100, 8320, 8325/8325P/8325H, 8360, 8400, 9300/9300S and 10000, disable routing by entering:

```
switch(config-lag-if)# no routing
```

See the Command-Line Interface Guide for your switch and software version for more information about the no routing command.

> **NOTE**
>
> On the 4100i, 6000, 6100, and 6200 Switch Series, routing is not supported on physical interfaces.
>
> On the 6300 and 6400 Switch Series, routing is disabled by default.

For example:

4. Assign a native VLAN ID to a trunk interface on the LAG by entering:

```
switch(config-lag-if)# vlan trunk native <VLAN-ID>
switch(config-lag-if)# vlan trunk native 1
```

5. Use the following steps to add a maximum of 16 interfaces to the LAG:

   a. To assign an interface to the LAG:

   ```
   switch(config-lag-if)# interface <PORT-ID>
   ```

   To assign a range of interfaces to a LAG:

```
switch(config-lag-if)# interface <PORT-ID>-<PORT-ID>
```

For example:

```
switch(config-lag-if)# interface 1/1/1-1/1/4
```

See the Command-Line Interface Guide for your switch and software version for more information about the `interface <PORT-ID>` command.

b. Assign an ID to the LAG:

```
switch(config-if)# lag <ID>
```

For example:

```
switch(config-if-<1/1/1-1/1/4>)# lag 100
```

c. Set the administrative state of the member interface to up:

```
switch(config-if-<1/1/1-1/1/4>)# no shutdown
```

6. View the configuration by entering the following:

For 4100i, 5420, 6000, 6100, 6200, 6300, and 6400 switch series:

```
switch(config-if-<1/1/1-1/1/4>)# show running-config
Current configuration:
!
vlan 1
interface lag 100
    no shutdown
    vlan trunk native 1
    vlan trunk allowed all
interface 1/1/1
    no shutdown
    lag 100
interface 1/1/2
    no shutdown
    lag 100
interface 1/1/3
    no shutdown
    lag 100
interface 1/1/4
    no shutdown
    lag 100
switch(config-if-<1/1/1-1/1/4>)# show lacp aggregates
Aggregate name    : lag100
```

```
Interfaces        : 1/1/3 1/1/1 1/1/4 1/1/2
Heartbeat rate    : N/A
Hash              : l3-src-dst
Aggregate mode    : Off
```

For 4100i, 5420, 6000, 6100, 6200, 6300, and 6400 switch series:

```
switch(config-if-<1/1/1-1/1/4>)# show running-config
Current configuration:
!
vlan 1
interface lag 100
    no shutdown
    vlan trunk native 1
    vlan trunk allowed all
interface 1/1/1
    no shutdown
    lag 100
interface 1/1/2
    no shutdown
    lag 100
interface 1/1/3
    no shutdown
    lag 100
interface 1/1/4
    no shutdown
    lag 100
switch(config-if-<1/1/1-1/1/4>)# show lacp aggregates
Aggregate name    : lag100
Interfaces        : 1/1/3 1/1/1 1/1/4 1/1/2
Heartbeat rate    : N/A
Hash              : l3-src-dst
Aggregate mode    : Off
```

For 8100, 8320, 8325/8325P/8325H, 8360, 8400, 9300/9300S and 10000 switch series:

```
switch(config-if-<1/1/1-1/1/4>)# show running-config
Current configuration:
!
vlan 1
interface lag 100
    no shutdown
    no routing
    vlan trunk native 1
    vlan trunk allowed all
interface 1/1/1
```

```
    no shutdown
    lag 100
interface 1/1/2
    no shutdown
    lag 100
interface 1/1/3
    no shutdown
    lag 100
interface 1/1/4
    no shutdown
    lag 100
switch(config-if-<1/1/1-1/1/4>)# show lacp aggregates
Aggregate name    : lag100
Interfaces        : 1/1/3 1/1/1 1/1/4 1/1/2
Heartbeat rate    : N/A
Hash              : l3-src-dst
Aggregate mode    : Off
```

Configuring a layer 2 dynamic aggregation group

Prerequisites

You must be in the global configuration context: `switch(config)#`.

Procedure

1. Create a layer 2 aggregate interface and access the layer 2 aggregate interface view by entering:

   ```
   switch(config)# interface lag <ID>
   ```

   While creating the layer 2 aggregate interface, the system automatically creates a layer 2 dynamic aggregation group numbered the same.

   The range of the LAG interface ID is 1 to 256.

2. Set the operational state of every interface in the LAG to up by entering:

```
switch(config-lag-if)# no shutdown
```

> **NOTE**
>
> This command does not impact the administrative state of the member interfaces because the command was entered at the level of the LAG. To change the administrative state of a member interface, enter the command at the interface level. For example:
>
> ```
> switch(config)# interface 1/1/2
> switch(config-if)# no shutdown
> ```

3. On the 8100, 8320, 8325/8325P/8325H, 8360, 8400, 9300/9300S and 10000, disable routing by entering:

   ```
   switch(config-lag-if)# no routing
   ```

   See the Command-Line Interface Guide for your switch and software version for more information about the `no routing` command.

   > **NOTE**
   >
   > On the 4100i, 6000, 6100, and 6200 Switch Series, routing is not supported on physical interfaces.
   >
   > On the 6300 and 6400 Switch Series, routing is disabled by default.

4. Configure the aggregation group to operate in dynamic mode by entering:

   ```
   switch(config-lag-if)# lacp mode {active | passive}
   ```

   For example:

   ```
   switch(config-lag-if)# lacp mode active
   ```

5. Configure the aggregation group to operate in fast or slow mode by entering:

   ```
   switch(config-lag-if)# lacp rate {fast | slow}
   ```

   For example:

   ```
   switch(config-lag-if)# lacp rate fast
   ```

6. Assign a native VLAN ID to a trunk interface by entering:

   ```
   switch(config-lag-if)# vlan trunk native <VLAN-ID>
   ```

For example:

```
switch(config-lag-if)# vlan trunk native 1
```

7. Use the following steps to add a maximum of 16 interfaces to the LAG:

   a. To assign an interface to the LAG:

   ```
   switch(config-lag-if)# interface <PORT-ID>
   ```

   To assign a range of interfaces to a LAG:

   ```
   switch(config-lag-if)# interface <PORT-ID>-<PORT-ID>
   ```

   For example:

   ```
   switch(config-lag-if)# interface 1/1/1-1/1/4
   ```

   See the Command-Line Interface Guide for your switch and software version for more information about the `interface <PORT-ID>` command.

   b. Assign an ID to the LAG:

   ```
   switch(config-if)# lag <ID>
   ```

   For example:

   ```
   switch(config-if-<1/1/1-1/1/4>)# lag 20
   ```

   c. Set the administrative state of the member interface to up:

   ```
   switch(config-if-<1/1/1-1/1/4>)# no shutdown
   ```

8. View the configuration by entering:
   For 4100i, 5420, 6000, 6100, 6200, 6300, and 6400 switch series:

```
switch(config-if-<1/1/1-1/1/4>)# show running-config
Current configuration:
!
vlan 1
interface lag 20
    no shutdown
    vlan trunk native 1
    vlan trunk allowed all
    lacp mode active
    lacp rate fast
interface 1/1/1
```

```
    no shutdown
    lag 20
switch(config-if-<1/1/1-1/1/4>)# show lacp aggregates
Aggregate name    : lag100
Interfaces        : 1/1/3 1/1/1 1/1/4 1/1/2
Heartbeat rate    : Fast
Hash              : l3-src-dst
Aggregate mode    : Active
```

For 8100, 8320, 8325/8325P/8325H, 8360, 8400, 9300/9300S and 10000 switch series:

```
switch(config-if-<1/1/1-1/1/4>)# show running-config
Current configuration:
!
vlan 1
interface lag 20
    no shutdown
    no routing
    vlan trunk native 1
    vlan trunk allowed all
    lacp mode active
    lacp rate fast
interface 1/1/1
    no shutdown
    lag 20
switch(config-if-<1/1/1-1/1/4>)# show lacp aggregates
Aggregate name    : lag100
Interfaces        : 1/1/3 1/1/1 1/1/4 1/1/2
Heartbeat rate    : Fast
Hash              : l3-src-dst
Aggregate mode    : Active
```

Layer 3 aggregation groups

Layer 3 aggregation groups are supported on all switch series except 6000, and 6100 Switch Series.

Subtopics

[Configuring a layer 3 static aggregation group]()
[Configuring a layer 3 dynamic aggregation group]()

Configuring a layer 3 static aggregation group

Prerequisites

You must be in the global configuration context: `switch(config)#`.

Procedure

1. Create a layer 3 aggregate interface and access the layer 3 aggregate interface view by entering:
   ```
   switch(config)# interface lag <ID>
   ```
   The range of the LAG interface ID is 1 to 256.

   While creating the layer 3 aggregate interface, the system automatically creates a layer 3 static aggregation group numbered the same.

2. Set the operational state of every interface in the LAG to up by entering:

   - For 5420, 6200, 6300 and 6400 switch series:
     ```
     switch(config-lag-if)# no shutdown
     switch(config-lag-if)# routing
     ```

     NOTE

     This command does not impact the administrative state of the member interfaces because the command was entered at the level of the LAG. To change the administrative state of a member interface, enter the command at the interface level. For example:

     ```
     switch(config)# interface 1/1/2
     switch(config-if)# no shutdown
     switch(config-if)# routing
     ```

   - For 8100, 8320, 8325/8325P/8325H, 8360, 8400, 9300/9300S and 10000 switch series:

```
switch(config-lag-if)# no shutdown
```

NOTE

This command does not impact the administrative state of the member interfaces because the command was entered at the level of the LAG. To change the administrative state of a member interface, enter the command at the interface level. For example:

```
switch(config)# interface 1/1/2
switch(config-if)# no shutdown
```

3. Set the IP address on the LAG interface by entering:

   ```
   switch(config-lag-if)# ip address <IPV4-ADDR>/<MASK>
   ```

   For example:

   ```
   switch(config-lag-if)# ip address 192.0.2.1/30
   ```

4. Use the following steps to add a maximum of 16 interfaces to the LAG:

   a. To assign an interface to the LAG:

      ```
      switch(config-lag-if)# interface <PORT-ID>
      ```

      To assign a range of interfaces to a LAG:

      ```
      switch(config-lag-if)# interface <PORT-ID>-<PORT-ID>
      ```

      For example:

      ```
      switch(config-lag-if)# interface 1/1/1-1/1/4
      ```

      See the Command-Line Interface Guide for your switch and software version for more information about the `interface <PORT-ID>` command.

   b. Assign an ID to the LAG:

      ```
      switch(config-if)# lag <ID>
      ```

      For example:

      ```
      switch(config-if-<1/1/1-1/1/4>)# lag 100
      ```

   c. Set the administrative state of the member interface to up:

```
switch(config-if-<1/1/1-1/1/4>)# no shutdown
```

5. View the configuration by entering the following:

   For 5420, 6200, 6300 and 6400 switch series:

```
switch(config-if-<1/1/1-1/1/4>)# show running-config
Current configuration:
!
vlan 1
interface lag 100
    no shutdown
    routing
    ip address 192.0.2.1/30
interface 1/1/1
    no shutdown
    lag 100
interface 1/1/2
    no shutdown
    lag 100
interface 1/1/3
    no shutdown
    lag 100
interface 1/1/4
    no shutdown
    lag 100
switch(config-if-<1/1/1-1/1/4>)# show lacp aggregates
Aggregate name   : lag100
Interfaces       : 1/1/3 1/1/1 1/1/4 1/1/2
Heartbeat rate   : N/A
Hash             : l3-src-dst
Aggregate mode   : Off
```

   For 8100, 8320, 8325/8325P/8325H, 8360, 8400, 9300/9300S and 10000 switch series:

```
switch(config-if-<1/1/1-1/1/4>)# show running-config
Current configuration:
!
vlan 1
interface lag 100
    no shutdown
    ip address 192.0.2.1/30
interface 1/1/1
    no shutdown
    lag 100
interface 1/1/2
```

```
    no shutdown
    lag 100
interface 1/1/3
    no shutdown
    lag 100
interface 1/1/4
    no shutdown
    lag 100
switch(config-if-<1/1/1-1/1/4>)# show lacp aggregates
Aggregate name    : lag100
Interfaces        : 1/1/3 1/1/1 1/1/4 1/1/2
Heartbeat rate    : N/A
Hash              : l3-src-dst
Aggregate mode    : Off
```

Configuring a layer 3 dynamic aggregation group

Prerequisites

You must be in the global configuration context: **switch(config)#**.

Procedure

1. Create a layer 3 aggregate interface and access the layer 3 aggregate interface view by entering:
   ```
   switch(config)# interface lag <ID>
   ```
   The range of the LAG interface ID is 1 to 256.

   While creating the layer 3 aggregate interface, the system automatically creates a layer 3 dynamic aggregation group numbered the same.
2. Set the operational state of every interface in the LAG to up by entering:
   - For 5420, 6200, 6300 and 6400 switch series:

```
switch(config-lag-if)# no shutdown
switch(config-lag-if)# routing
```

NOTE

This command does not impact the administrative state of the member interfaces because the command was entered at the level of the LAG. To change the administrative state of a member interface, enter the command at the interface level. For example:

```
switch(config)# interface 1/1/2
switch(config-if)# no shutdown
switch(config-if)# routing
```

- For 8100, 8320, 8325/8325P/8325H, 8360, 8400, 9300/9300S and 10000 switch series:

```
switch(config-lag-if)# no shutdown
```

NOTE

This command does not impact the administrative state of the member interfaces because the command was entered at the level of the LAG. To change the administrative state of a member interface, enter the command at the interface level. For example:

```
switch(config)# interface 1/1/2
switch(config-if)# no shutdown
```

3. Configure the aggregation group to operate in dynamic mode by entering:

```
switch(config-lag-if)# lacp mode {active | passive}
```

For example:

```
switch(config-lag-if)# lacp mode active
```

4. Configure the aggregation group to operate in fast or slow mode by entering:

```
switch(config-lag-if)# lacp rate {fast | slow}
```

For example:

```
switch(config-lag-if)# lacp rate fast
```

5. Set the IP address on the LAG interface by entering:

```
switch(config-lag-if)# ip address <IPV4-ADDR>/<MASK>
```

For example:

```
switch(config-lag-if)# ip address 192.0.3.1/30
```

6. Use the following steps to add a maximum of 16 interfaces to the LAG:

   a. To assign an interface to the LAG:

   ```
   switch(config-lag-if)# interface <PORT-ID>
   ```

   To assign a range of interfaces to a LAG:

   ```
   switch(config-lag-if)# interface <PORT-ID>-<PORT-ID>
   ```

   For example:

   ```
   switch(config-lag-if)# interface 1/1/1-1/1/4
   ```

   See the Command-Line Interface Guide for your switch and software version for more information about the `interface <PORT-ID>` command.

   b. Assign an ID to the LAG:

   ```
   switch(config-if)# lag <ID>
   ```

   For example:

   ```
   switch(-<1/1/1-1/1/4>)# lag 100
   ```

   c. Set the administrative state of the member interface to up:

   ```
   switch(-<1/1/1-1/1/4>)# no shutdown
   ```

7. View the configuration by entering:
   For 5420, 6200, 6300 and 6400 switch series:

```
switch(config-if-<1/1/1-1/1/4>)# show running-config
Current configuration:
!
vlan 1
interface lag 100
    no shutdown
    routing
    ip address 192.0.3.1/30
    lacp mode active
```

```
    lacp rate fast
interface 1/1/1
    no shutdown
    lag 100
interface 1/1/2
    no shutdown
    lag 100
interface 1/1/3
    no shutdown
    lag 100
interface 1/1/4
    no shutdown
    lag 100
switch(config-if-<1/1/1-1/1/4>)# show lacp aggregates
Aggregate name   : lag100
Interfaces       : 1/1/3 1/1/1 1/1/4 1/1/2
Heartbeat rate   : Fast
Hash             : l3-src-dst
Aggregate mode   : Active
```

For 8100, 8320, 8325/8325P/8325H, 8360, 8400, 9300/9300S and 10000 switch series:

```
switch(config-if-<1/1/1-1/1/4>)# show running-config
Current configuration:
!
vlan 1
interface lag 100
    no shutdown
    ip address 192.0.3.1/30
    lacp mode active
    lacp rate fast
interface 1/1/1
    no shutdown
    lag 100
interface 1/1/2
    no shutdown
    lag 100
interface 1/1/3
    no shutdown
    lag 100
interface 1/1/4
    no shutdown
    lag 100
switch(config-if-<1/1/1-1/1/4>)# show lacp aggregates
Aggregate name   : lag100
```

```
Interfaces         : 1/1/3 1/1/1 1/1/4 1/1/2
Heartbeat rate     : Fast
Hash               : l3-src-dst
Aggregate mode     : Active
```

Removing a LAG

Prerequisites

You must be in the global configuration context: `switch(config)#`.

Procedure

Delete the LAG. Enter:

```
switch(config)# no interface lag <ID>
```

For example:

```
switch(config)# no interface lag 100
```

All interfaces assigned to the LAG are automatically removed from the LAG as part of the deletion process of the LAG. After removing a physical interface from a LAG,

- **4100i, 5420, 6000, 6100, 6200, 6300, and 6400 switches:** The interface associated with the LAG becomes layer 2 ports with the default layer 2 configurations and **admin** status enabled.
- **8100, 8320, 8235, 8360, and 8400 switches:** The interface associated with the LAG becomes layer 3 ports with default layer 3 configurations and administrative down.

Removing an interface from a LAG

Prerequisites

You must be in the global configuration context: `switch(config)#`.

Procedure

Remove an interface from the LAG. Enter:

```
switch(config)# interface <PORT-NUM>
switch(config-if)# no lag <ID>
```

For example:

```
switch(config)# interface 1/1/1
switch(config-if)# no lag 100
switch(config-if)# show running-config
...
!
vlan 1
interface lag 100
interface 1/1/1
interface 1/1/2
    lag 100
switch(config-if)#
```

To assign a range of interfaces to a LAG:

```
switch(config-lag-if)# interface <PORT-ID>-<PORT-ID>
```

For example:

```
switch(config-lag-if)# interface 1/1/1-1/1/4
```

After removing a physical interface from a LAG:

- **4100i, 5420, 6000, 6100, 6200, 6300, and 6400 switches:** The interface associated with LAG becomes layer 2 ports with default layer 2 configurations and with admin status of **enabled**
- **8100, 8320, 8325/8325H/8325P, 8360, 8400, 9300/9300S and 10000 switches:** The interface associated with the LAG becomes L3 ports with default L3 configurations and administrative down. For example, suppose interface 1/1/1 was part of LAG 3 and you had administratively enabled the interface. If you later remove interface 1/1/1 from LAG 3, the administrative status automatically changes to down. If you want to use the interface again, you must administratively enable it again.

Changing the LAG membership for an interface

Prerequisites

You must be in the global configuration context: `switch(config)#`.

Procedure

1. Remove an interface from the LAG. Enter:

   ```
   switch(config)# interface <PORT-NUM>
   switch(config-if)# no lag <ID>
   ```

   For example:

   For 4100i, 5420, 6000, 6100, 6200, 6300, and 6400 switch series:

```
switch(config)# interface 1/1/1
switch(config-if)# no lag 100
switch(config-if)# show running-config
Current configuration:
!
...
!
vlan 1
interface lag 100
    no shutdown
    vlan trunk native 1
    vlan trunk allowed all
interface 1/1/1
    no shutdown
interface 1/1/2
    no shutdown
    lag 100
switch(config-if)#
```

For 8100, 8320, 8325/8325P/8325H, 8360, 8400, 9300/9300S and 10000 switch series:

```
switch(config)# interface 1/1/1
switch(config-if)# no lag 100
switch(config-if)# show running-config
Current configuration:
!
...
!
vlan 1
interface lag 100
    no shutdown
    no routing
    vlan trunk native 1
    vlan trunk allowed all
interface 1/1/1
interface 1/1/2
    no shutdown
    no routing
   vlan access 1
switch(config-if)#
```

After removing a physical interface from a LAG, the interface associated with the LAG becomes L3 ports with default L3 configurations and administrative down. For example, suppose interface 1/1/1 was part of LAG 3 and you had administratively enabled the interface. If you later remove interface 1/1/1 from LAG 3, the administrative status automatically changes to down.

On 4100i, 5420, 6000, 6100, and 6200 Switch Series, after removing a physical interface from a LAG, the interface associated with the LAG becomes layer 2 ports with default layer 2 configurations and **admin** status enabled.

2. Create the LAG to which you want to add the interface:

```
switch(config-if)# interface lag 10
```

For 4100i, 5420, 6000, 6100, 6200, 6300, and 6400 switch series:

```
switch(config-if)# interface lag 10
switch(config-lag-if)# no shutdown
switch(config-lag-if)# vlan trunk native 1
```

For example:

For 8100, 8320, 8325/8325P/8325H, 8360, 8400, 930/9300Sand 10000 switch series:

switch(config-if)# **interface lag 10** switch(config-lag-if)# **no shutdown** switch(config-lag-if)# **no routing** switch(config-lag-if)# **vlan trunk native 1**

3. Add the interface from Step 1 to the newly created LAG:

```
switch(config)# interface 1/1/1
switch(config-if)# lag 10
```

For example:

For 4100i, 5420, 6000, 6100, 6200, 6300, and 6400 switch series:

```
switch(config)# interface 1/1/1
switch(config-if)# lag 10
switch(config-if)# show running-config
Current configuration:
!
...
!
vlan 1
interface lag 10
    no shutdown
    vlan trunk native 1
    vlan trunk allowed all
interface lag 100
    no shutdown
    vlan trunk native 1
    vlan trunk allowed all
interface 1/1/1
    lag 10
```

```
interface 1/1/2
    no shutdown
    lag 100
```

For 8100, 8320, 8325/8325P/8325H, 8360, 8400, 930/9300Sand 10000 switch series:

```
switch(config)# interface 1/1/1
switch(config-if)# lag 10
switch(config-if)# show running-config
Current configuration:
!
...
!
vlan 1
interface lag 10
    no shutdown
    no routing
    vlan trunk native 1
    vlan trunk allowed all
interface lag 100
    no shutdown
    no routing
    vlan trunk native 1
    vlan trunk allowed all
interface 1/1/1
    lag 10
interface 1/1/2
    no shutdown
    lag 100
```

The interface is enabled by default when it is added to the LAG.

Configuration of an aggregate Interface

Subtopics

Configuring the description of an aggregate interface

You can configure the description of an aggregate interface for administration purposes, for example, describing the purpose of the interface.

Prerequisites

You must be in the global configuration context: `switch(config)#`.

Procedure

1. Create a layer 3 aggregate interface and enter layer 3 aggregate interface view by entering:
   ```
   switch(config)# interface lag <ID>
   ```
2. Configure the description of the aggregate interface:
   ```
   switch(config-if)# description <text>
   ```

Setting the MTU for a layer 2 member link interface

Prerequisites

You must be in the global configuration context: `switch(config)#`.

Procedure

1. Enter a layer 2 member link interface view by entering:
   ```
   switch(config)# interface <INTERFACE-ID>
   ```
2. Set the MTU for the layer 2 member link interface:
   ```
   switch(config-if)# mtu <VALUE>
   ```
   See the Command-Line Interface Guide for your switch and software version for more information about the mtu <VALUE> command. When allowing jumbo frames under a layer 2 aggregation interface, make sure that the MTU value is set appropriately under all member interfaces.

Setting the MTU for a layer 3 aggregate interface

> **NOTE**
>
> Layer 3 aggregation groups are not supported on the 4100i, 5420, 6000, 6100, and 6200 Switch Series.

The MTU of an interface affects IP packets fragmentation and reassembly on the interface.

Prerequisites

You must be in the global configuration context: `switch(config)#`.

Procedure

1. Enter layer 3 aggregate interface view by entering:
   ```
   switch(config)# interface lag <INTERFACE-ID>
   ```
2. Set the MTU for the layer 3 aggregate interface:
   ```
   switch(config-lag-if)# ip mtu <VALUE>
   ```
   See the Command-Line Interface Guide for your switch and software version for more information about the ip mtu <VALUE> command. When allowing jumbo frames under a layer 2 aggregation interface, make sure that the MTU value is set appropriately under all member interfaces.

   > **NOTE**
   >
   > If the IP MTU is configured as 9198, the MTU on the physical interfaces must also be configured as 9198.

Impact of shutting down or bringing up an aggregate interface

By default, an aggregate interface is down. Shutting down or bringing up an aggregate interface affects the aggregation states and link states of member ports in the corresponding aggregation group as follows:

- When an aggregate interface is shut down, all Selected ports in the corresponding aggregation group become Unselected ports and all member ports go to an operationally down state.
- When an aggregate interface is brought up, the aggregation states of member ports in the corresponding aggregation group are recalculated. LAG members, which are administratively up, will become operationally up. The members that are not administratively up will be in the same state and not made eligible for aggregation.

Shutting down an aggregate interface

Prerequisites

You must be in the global configuration context: `switch(config)#`.

Procedure

Enter the layer 3 aggregate interface view by entering:

```
switch(config)# interface lag <ID>
```

Shut down the aggregate interface:

```
switch(config-lag-if)# shutdown
```

Supported hashing algorithms

- Source MAC and destination MAC
- Source IP and destination IP
- Source port and destination port.

Configuration verification

Task	Command	Example

Viewing LACP global information	`show lacp configuration`	`switch# **show lacp configuration**`
		`System-id : 70:72:cf:ef:fc:d9`
		`System-priority : 65534`
		`Hash : l3-src-dst`

Task	Command	Example
		The output displayed for the `show lacp configuration` is from the 8400 series switch.

Viewing LACP aggregate information	`show lacp aggregates`	`switch# **show lacp aggregates**`
		`Aggregate-name : lag100`
		`Aggregated-interfaces : 1/1/2`
		`Heartbeat rate : N/A`
		`Hash : l3-src-dst`
		`Aggregate mode : off`
		`Aggregate-name : lag110`
		`Aggregated-interfaces : 1/1/1, 1/1/3`
		`Heartbeat rate : slow`
		`Hash : l3-src-dst`
		`Aggregate mode : active`

Viewing LACP aggregate information for a LAG	`show lacp aggregates lag100`	`switch# **show lacp aggregates lag100**`
		`Aggregate-name : lag100`
		`Aggregated-interfaces :`
		`Heartbeat rate : N/A`
		`Hash : l3-src-dst`
		`Aggregate mode : off`

Viewing LACP interface details	`show lacp interfaces`	`switch# **show lacp interfaces**`

Task	Command	Example
		The output is too wide to display in a column. The command output is provided in the CLI topic for the command.

BFD reports a LAG as down even when healthy links are still available

Symptom

NOTE

BFD is not supported on the 4100i, 5420, 6000, 6100, and 6200 Switch Series.

The Bidirectional Forward Detection (BFD) feature reports a Link Aggregation (LAG), as being down, even though there are healthy LAG links available. The LAG, containing the downed link, will eventually rebalance the traffic to its other links.

Cause

This notification occurs when the minimum BFD control packet reception interval is set at a faster rate than the Link Aggregation Control Protocol (LACP) rate and LAG rebalancing occurs. BFD assumes that the link is down without realizing that the LAG is rebalancing the traffic load.

Action

Set the minimum BFD control packet reception interval to a slower rate than the LACP rate or set the LACP rate to a faster rate than the minimum BFD control packet reception interval.

1. To find the current settings of the minimum BFD control packet reception interval, enter the `show running-config` command.
   The minimum BFD control packet reception interval setting is listed as bfd min-receive-interval in the command output and the measurement is in ms.
2. To find the current rate of LACP, enter the `show lacp aggregates` command.
   The LACP rate is listed as the Heatbeat rate in the command output.
3. To change the minimum BFD control packet reception interval, enter the `bfd min-receive-interval` command.
4. To change the LACP rate, enter the `lacp rate {fast | slow}` command.

LACP and LAG commands

Select a command from the list in the left navigation menu.

Subtopics

`description`

Syntax

```
description <TEXT>
no description <TEXT>
```

Description

Provides a brief description of the LAG interface. The description text is saved in the configuration of the LAG. It is available even after a reboot.

The no form of this command removes the description of the LAG interface from the configuration.

Parameter	Description
<TEXT>	Specifies the description of the LAG interface.

Example

```
switch(config)# interface lag 10
switch(config-lag-if)# description This LAG is used for an example.
switch(config-lag-if)# show running-config
...
vlan 1
interface lag 10
    description This LAG is used for an example.
interface lag 60
switch(config-lag-if)#
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	`config-lag-if`	Administrators or local user group members with execution righ ts for this command.

`hash`

Syntax

```
hash [l2-src-dst | l3-src-dst | l4-src-dst]
```

Description

This command controls the selection of an interface in a group of aggregate interfaces. The hash type value helps transmit a frame. This configuration must be done at the LAG interface level.

Parameter	Description
`l2-src-dst`	Specifies the load - balancing calculation to include only layer 2 items, such as source and destination MAC addresses.
`l3-src-dst`	Specifies the load - balancing calculation to include only layer 3 items, such as source and destination IP addresses. Default s etting.
`l4-src-dst`	Specifies the load - balancing calculation to include only layer 4 items, such as source and destination UDP/TCP ports.

Example

```
switch(config-lag-if)# hash l2-src-dst
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200 6300 6400 8100 8320 8325 8325H 8325P	`config-lag-if`	Administrators or local user group members with execution righ ts for this command.

Platforms	Command context	Authority
10000		

`interface lag`

Syntax

```
interface lag <ID>
no interface lag <ID>
```

Description

Creates a Link Aggregation Group (LAG) interface represented by an ID.

The **no** form of this command deletes a LAG interface represented by an ID.

Parameter	Description
<ID>	Specifies a LAG interface ID.

Usage

Keep in mind the following requirements when adding interfaces to a LAG:

- To determine the maximum number of LAG interfaces for your type of switch, look at the output from the **show capacities lag** command; however, the number of LAGs that can be created depends on the availability of the physical interface since each LAG interface needs at least one physical interface as a member link.
- After the maximum limit of members is reached in a LAG, an additional port cannot be added to the aggregation group. If a port belongs to a card type with a different speed than the other aggregation members, the port can still be added to the aggregation group. If dynamic LAG is enabled, any port member with a speed different than other aggregation members is blocked or ineligible from the same aggregation group. Any operational keys/attributes or configuration changes might affect the aggregation states of the member ports.
- The nondefaults configuration on an interface is removed automatically when the interface is added to a link aggregation. For example: Assume that you remove a member interface from an existing LAG and

add it to another LAG. The software removes the nondefault configurations on the interface when it is added to the new LAG.

Examples

Creating a Link Aggregation Group (LAG) interface represented by an ID of 100:

```
switch(config)# interface lag 100
```

Deleting a Link Aggregation Group (LAG) interface represented by an ID of 100:

```
switch(config)# no interface lag 100
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution righ ts for this command.

`interface persona lag`

Syntax

```
interface persona lag <NAME>
no interface persona lag <NAME>
```

Description

Creates a persona LAG, a template for LAG (Link Aggregation Group) interfaces. A persona LAG defines configuration options that can be applied to multiple LAGs for consistency and simplified management. Only LAG interfaces can reference persona LAG templates. The persona LAG name cannot use reserved interface names or formats, such as vlan, lag, or loopback. Configurations defined in a persona LAG are valid only for LAG interfaces.

The **no** form of this command deletes the persona LAG interface.

Parameter	Description
<NAME>	Specifies the name of the persona LAG.

Usage

Keep in mind the following requirements when adding interfaces to a LAG:

- Persona LAG names must not match reserved interface names or formats (e.g., vlan, lag, loopback).
- Only LAG interfaces can use persona LAGs.
- Configuration applied to a persona LAG is only applicable to LAG interfaces.
- The persona LAG name is case-sensitive and can contain up to 54 characters.
- Names can include any characters, but avoid spaces or characters that may interfere with CLI parsing.
- Multi-chassis LAG interfaces do not support personas.

Examples

Creating a persona LAG template:

```
switch(config)# interface persona lag server-uplink
switch(config-persona-lag)# no shutdown
switch(config-persona-lag)# mtu 9000
switch(config-persona-lag)# description "Server uplink template"
```

Deleting a persona LAG template:

```
switch(config)# no interface persona lag server-uplink
```

Command History

Release	Modification
10.17.1000	Command introduced on 10040 Switch Series.
10.17	Command introduced.

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution righ ts for this command.

`ip address`

Syntax

```
ip address <IPV4-ADDR>/<MASK> [secondary]
no ip address <IPV4-ADDR>/<MASK> [secondary]
```

Description

Sets an IPv4 address and subnet mask to a LAG interface. One primary and up to 31 secondary address can be configured per interface.

The **no** form of this command removes the IPv4 address from the interface.

Parameter	Description
<IPV4-ADDR>	Specifies an IP address in IPv4 format (**x.x.x.x**), where **x** is a de cimal number from 0 to 255. You can remove leading zeros. For example, the address **192.169.005.100** becomes **192.168.5.1 00**.
<MASK>	Specifies the number of bits in the address mask in CIDR format (**x**), where **x** is a decimal number from 0 to 32.
`secondary`	Specifies a secondary IP address.

Examples

Setting an IP address on the LAG interface 1 to 198.51.100.1 with a mask of 24 bits:

```
switch(config)# interface lag 1
switch(config-lag-if)# ip address 198.51.100.1/24
```

Removing the IP address 198.51.100.1 with a mask of 24 bits from LAG interface 1:

```
switch(config)# interface lag 1
switch(config-lag-if)# no ip address 198.51.100.1/24
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420 6200 6300 6400 8100 8320 8325 8325H 8325P 8360 8400 9300 9300S 10000 10040	`config-lag-if`	Administrators or local user group members with execution righ ts for this command.

`ipv6 address`

Syntax

```
ipv6 address <IPV6-ADDR>/<MASK>
no ipv6 address <IPV6-ADDR>/<MASK>
```

Description

Sets an IPv6 address and subnet mask to a LAG interface.

The **no** form of this command removes the IPv6 address from the interface.

Parameter	Description
<IPV6-ADDR>	Specifies the IP address in IPv6 format (**xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx**), where **x** is a hexadecimal number from 0 to F. You can use two colons (::) to represent consecutive zeros (but only once), remove leading zeros, and collapse a quartet of four zeros to a single 0. For example, this address **2222:0000:3333:0000:0000:0000:4444:0055** becomes **2222:0:3333::4444:55**.
<MASK>	Specifies the number of bits in the address mask in CIDR format (**x**), where **x** is a decimal number from 0 to 128.

Examples

Setting the IPv6 address on LAG interface 1 to 2001:0db8:85a3::8a2e:0370:7334 with a mask of 24 bits:

```
switch(config)# interface lag 1
switch(config-lag-if)# ipv6 address 2001:0db8:85a3::8a2e:0370:7334/24
```

Removing the IP address 2001:0db8:85a3::8a2e:0370:7334 with mask of 24 bits with a mask of 24 bits from LAG interface 1:

```
switch(config)# interface lag 1
switch(config-lag-if)# no ipv6 address 2001:0db8:85a3::8a2e:0370:7334/24
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200 6300 6400 8100 8320 8325 8325H 8325P 8360 8400 9300 9300S 10000 10040	`config-lag-if`	Administrators or local user group members with execution righ ts for this command.

`lacp fallback`

Syntax

```
lacp fallback
no lacp fallback
```

Description

Configures the LACP fallback on LAG port.

The no form of this command sets the LAG to BLOCK state if no LACP partner is detected.

Usage

This makes members of the LAG function as non-bonded interfaces when no LACP partner is detected. This configuration is only applicable when the LAG is of type MCLAG. If the member port does not get an LACP frame, the port is in IE state.

Examples

Configuring LACP fallback on LAG port.

```
switch(config)# int lag 1 multi-chassis
switch(config-lag-if)# no sh
switch(config-lag-if)# lacp mode active
switch(config-lag-if)# lacp fallback
```

Configuring the LAG to BLOCK state when no LACP partner is detected.

```
switch(config)# int lag 1 multi-chassis
switch(config-lag-if)# no sh
switch(config-lag-if)# lacp mode active
switch(config-lag-if)# no lacp fallback
```

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200 6400 8100 8320 8325 8325H 8325P 8400 9300 9300S 10000 10040	`config-if` `config-lag-if`	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

lacp fallback-static

Syntax

```
lacp fallback-static
no lacp fallback-static
```

Description

Configures the LACP fallback-static on LAG port.

The no form of this command sets the LAG to BLOCK state if no LACP partner is detected.

Usage

This makes members of the LAG function as non-bonded interfaces when no LACP partner is detected. One member interface that is part of the LAG stays up and forwards traffic, while the other members are in lacp-block state. This configuration is applicable when the lag is of type LACP and ignored in other cases. When this command is configured, only one member of LAG is selected to be UP. Enabling multiple members results in configuration mismatch on peer, loop, mac-learning issues, and more.

Examples

Configuring LACP fallback-static on LAG port.

```
switch(config)# interface lag 1
switch(config-lag-if)# lacp mode active
switch(config-lag-if)# lacp fallback-static
```

Configuring the LAG to BLOCK state when no LACP partner is detected.

```
switch(config)# interface lag 1
switch(config-lag-if)# no lacp fallback-static
```

Configuring LACP fallback-static on static port.

```
switch(config-lag-if)# lacp fallback-static
Cannot enable LACP fallback-static on static LAG.
```

Release	Modification
10.11	Command introduced.

Command Information

Platforms	Command context	Authority
All platforms	`config-if` `config-lag-if`	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

`lacp hash`

Syntax

```
lacp hash [l2-src-dst | l3-src-dst | l4-src-dst]
no lacp hash [l2-src-dst | l3-src-dst | l4-src-dst]
```

Description

Controls the selection of an interface in a group of aggregate interfaces. The hash type value helps transmit a frame. This configuration must be done at the global level.

Parameter	Description
`l2-src-dst`	Specifies the load - balancing calculation to include only layer 2 items, such as source and destination MAC addresses.
`l3-src-dst`	Specifies the load - balancing calculation to include only layer 3 items, such as source and destination IP addresses.
`l4-src-dst`	Specifies the load - balancing calculation to include only layer 4 items, such as source and destination UDP/TCP ports.

Example

```
switch(config)# lacp hash l2-src-dst
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
6000 6100 8400 9300	`config`	Administrators or local user group members with execution righ ts for this command.

`lacp mode`

Syntax

```
lacp mode {active | passive}
no lacp mode {active | passive}
```

Description

Sets an LACP mode to active or passive.

The **no** form of this command sets the LACP mode to **off**, returning the LAG to a static mode aggregation.

Parameter	Description
`active`	Specifies that the local switch will transmit LACP Data Units (L ACPDUs) to attempt to negotiate with the remote device.
`passive`	Specifies that the local switch will listen for LACPDUs from the remote device for LACP negotiation. **NOTE** A momentary traffic drop occurs because LACP partners reconverge when changing the mode f rom active to passive or from passive to active.

Examples

Setting the LACP mode to `active`:

```
switch(config)# interface lag 1
switch(config-lag-if)# lacp mode active
```

Setting the LACP mode to `off`:

```
switch(config)# interface lag 1
switch(config-lag-if)# no lacp mode active
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	`config-lag-if`	Administrators or local user group members with execution righ ts for this command.

`lacp port-id`

Syntax

```
lacp port-id <PORT-ID>
no lacp port-id
```

Description

Sets the LACP port ID value of the member interface of the LAG.

The **no** form of this command removes the LACP port ID value from the interface.

Parameter	Description
<PORT-ID>	Specifies a port ID value. Range: 1 to 65535.

Examples

Setting an LACP port ID to a value of 10:

```
switch(config-if)# lacp port-id 10
```

Removing the LACP port ID value:

```
switch(config-if)# no lacp port-id
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	`config-if`	Administrators or local user group members with execution righ ts for this command.

`lacp port-priority`

Syntax

```
lacp port-priority <PORT-PRIORITY>
no lacp port-priority
```

Description

Sets an LACP port priority value for the member interface of the LAG.

The **no** form of this command reverts the LACP port priority to the default, which is 1.

Parameter	Description
	Specifies a port priority value. Range: 1 to 65535.

Parameter	Description
<PORT-PRIORITY>	

Examples

Setting a LACP port priority value of 10:

```
switch(config-if)# lacp port-priority 10
```

Reverting the LACP port ID to the default:

```
switch(config-if)# no lacp port-priority
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	`config-if`	Administrators or local user group members with execution righ ts for this command.

`lacp rate`

Syntax

```
lacp rate {fast | slow}
no lacp rate {fast | slow}
```

Description

Sets an LACP heartbeat request time to fast or slow.

The **no** form of the command sets an LACP rate to **slow**.

Parameter	Description
`fast`	Specifies the heartbeat request to every second, and the timeout period is a three - consecutive heartbeat loss that is 3 seconds.
`slow`	Specifies the heartbeat request to every 30 seconds. The timeout period is three - consecutive heartbeat loss that is 90 seconds. Default setting.

Examples

Setting the LACP heartbeat request time to `fast`:

```
switch(config)# interface lag 1
switch(config-lag-if)# lacp rate fast
```

Resetting the LACP heartbeat request time to the default, which is `slow`:

```
switch(config)# interface lag 1
switch(config-lag-if)# no lacp rate
```

Another way to set the LACP heartbeat request time to the default, which is `slow`:

```
switch(config)# interface lag 1
switch(config-lag-if)# lacp rate slow
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	`config-lag-if`	Administrators or local user group members with execution rights for this command.

`lacp system-priority`

Syntax

```
lacp system-priority <SYSTEM-PRIORITY-VALUE>
no lacp system-priority <SYSTEM-PRIORITY-VALUE>
```

Description

Sets a Link Aggregation Control Protocol (LACP) system priority.

The **no** form of this command sets an LACP system priority to the default, which is 65534.

Parameter	Description
`<SYSTEM-PRIORITY-VALUE>`	Specifies a system priority value. Range: 0 to 65535.

Examples

Setting a Link Aggregation Control Protocol (LACP) system priority to 100:

```
switch(config)# lacp system-priority 100
```

Setting an LACP system priority to the default (65534):

```
switch(config)# no lacp system-priority
```

A momentary traffic drop can be seen in case the LACP state machine must renegotiate.

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution righ ts for this command.

`lag`

Syntax

```
lag <ID>
no lag <ID>
```

Description

Adds an interface to a specified LAG interface ID.

The **no** form of this command removes an interface from a specified LAG interface ID. The member loses its LACP configuration when removed from the LAG. The member also reaches the default state with an administrative shutdown. For 6300 and 6400 series switches, the administrative state is enabled. Configurations, such as MTU and UDLD, are retained.

Parameter	Description
`<ID>`	Specifies a LAG interface ID. Range: 1 to 256.

Usage

- All members of the LAG must have the same speed. If a member comes up late with a different speed, it will not participate in the LAG/LACP. The hardware restriction is applied before adding an interface to LAG. The member belongs to the card type that has the same maximum speed as the reference port card type.
- To move an interface from LagA to LagB, first remove the interface from LagA and then add it to LagB. When a member is attached to a LAG, the nondefault configurations on the member are removed silently.
- After removing a physical interface from a LAG, the interface associated with the LAG becomes L3 ports with default L3 configurations and administrative down. For example, suppose interface 1/1/1 was part of LAG 3 and you had administratively enabled the interface. If you later remove interface 1/1/1 from LAG 3, the administrative status automatically changes to down. If you want to use the interface again, you must administratively enable it again.

Examples

Adding an interface to a Link Aggregation Group (LAG) represented by an ID of 100:

```
switch(config)# interface 1/1/1
switch(config-if)# lag 100
```

Deleting an interface from a Link Aggregation Group (LAG) represented by an ID of 100:

```
switch(config)# interface 1/1/1
switch(config-if)# no lag 100
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	`config-if`	Administrators or local user group members with execution righ ts for this command.

`persona custom`

Syntax

```
persona custom <NAME> {copy | attach}
no persona custom <NAME> {copy | attach}
```

Description

Assigns a persona LAG (Link Aggregation Group) to a LAG interface. The persona LAG is used to apply a common configuration to multiple LAGs. There are two modes for applying a persona LAG: `copy` and `at tach`.

The **no** form of this command deletes the persona LAG.

Parameter	Description
	Specifies the custom interface persona name. Range: 1 to 54 ch aracters.

Parameter	Description
`<NAME>`	
`copy`	Performs a one - time copy of the template LAG configuration. Subsequent changes to the template are not applied. If the spec ified persona LAG does not exist, the system displays the messa ge `persona LAG <name> not found`.
`attach`	Performs an initial copy of the template LAG configuration, and any subsequent changes to the template are automatically appl ied to all attached LAGs. The template LAG does not need to ex ist before other LAGs are attached to it. After attachment, the c opied settings can be modified on an individual LAG. However, any later change to the template overwrites the modified values with the updated template values. Existing LAG configuration p rior to attachment is discarded and is not restored if the templ ate is removed.

Usage

Keep in mind the following when assigning a persona LAG:

- Only LAG interfaces can be assigned a persona LAG.
- Physical interfaces cannot be assigned a persona LAG.
- When using the `copy` option, the persona LAG must exist before it can be used; for the `attach` option, it is created if it does not already exist.
- If a persona LAG is deleted, all LAGs referencing it retain their last applied settings.
- LAG configuration that existed prior to attaching a persona LAG is discarded and is not restored if the template is removed.
- When using the `attach` option, changes to the persona LAG template are immediately applied to all attached LAGs.

Examples

Creating a persona LAG template and assigning it to a LAG with the `attach` option:

```
switch(config)# interface persona lag server-uplink
switch(config-persona-lag)# mtu 9000
switch(config-persona-lag)# description "Server uplink LAG"
switch(config-persona-lag)# exit
switch(config)# interface lag 10
switch(config-lag-if)# persona custom server-uplink attach
```

Creating a persona LAG template and assigning it to a LAG with the `copy` option:

```
switch(config)# interface persona lag server-uplink
switch(config-persona-lag)# mtu 9000
switch(config-persona-lag)# description "Server uplink LAG"
switch(config-persona-lag)# exit
switch(config)# interface lag 11
switch(config-lag-if)# persona custom server-uplink copy
switch(config-lag-if)# end
```

Removing a persona LAG from a LAG:

```
switch(config)# interface lag 11
switch(config-lag-if)# no persona custom server-uplink attach
```

Removing a persona LAG assignment completely:

```
switch(config)# interface lag 11
switch(config-lag-if)# no persona custom server-uplink
```

Command History

Release	Modification
10.17.1000	Command introduced on 10040 Switch Series.
10.17	Command introduced.

Command Information

Platforms	Command context	Authority
All platforms	`config-lag-if`	Administrators or local user group members with execution righ ts for this command.

`show interface`

Syntax

```
show interfaces <LAG-NAME>
             [vsx-peer]
```

Description

Displays information about a specific LAG.

Parameter	Description
`<LAG-NAME>`	Specifies a LAG name.
`vsx-peer`	Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output f rom the VSX peer switch is not displayed. This parameter is avai lable on switches that support VSX.

Examples

Displaying information about LAG 100:

```
switch# show interface lag100
Aggregate lag100 is up
 Admin state is up
 Description :
 MAC Address                   : 48:0f:cf:af:43:9c
 Aggregated-interfaces         : 1/1/2
 Aggregation-key               : 100
 Aggregate mode                : active
 Speed                         : 2000 Mb/s
 L3 Counters: Rx Disabled, Tx Disabled
 qos trust none
 VLAN Mode: access
 Access VLAN: 1
 Statistics                        RX                     TX
Total
 ------------- -------------------- --------------------
--------------------
 Packets                           20                     45
65
   Unicast                          5                      5
```

```
10
   Multicast                         5                    15
20
   Broadcast                        10                    25
35
 Bytes                            5658                  2584
8242
 Jumbos                              0                     0
0
 Dropped                             0                     0
0
 Filtered                            0                     0
0
 Pause Frames                        0                     0
0
 Errors                              0                     0
0
   CRC/FCS                           0                   n/a
0
   Collision                       n/a                     0
0
   Runts                             0                   n/a
0
   Giants                            0                   n/a
0
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Operator (`>`) or Manage r (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

show interface persona

Syntax

```
show interface persona
```

Description

Shows all interface personas and their attached interfaces.

Examples

Showing all interface personas and their attached interfaces:

```
switch(config)# show interface persona
-------------------------------------------------------
Persona              Type                 Attached Interfaces
-------------------------------------------------------
access               physical             1/1/1
persona1             physical             1/1/1-1/1/2
persona2             physical             --
access_lag           lag                  lag20,lag30
```

Command History

Release	Modification
10.17.1000	Command introduced on 10040 Switch Series.
10.17	Added support for persona LAG interfaces.
10.12	Command introduced.

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution righ ts for this command.

show lacp aggregates

Syntax

```
show lacp aggregates [<LAG-NAME>] [vsx-peer]
```

Description

Displays all LACP aggregate information configured for all LAGs, or for a specific LAG.

Parameter	Description
`<LAG-NAME>`	Optional: Specifies a lag name.
`vsx-peer`	Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output f rom the VSX peer switch is not displayed. This parameter is avai lable on switches that support VSX.

Examples

Displaying LACP aggregate information configured for lag10:

```
switch# show lacp aggregates lag10
Aggregate-name         : lag10
Aggregated-interfaces : 1/1/1 1/1/2
Heartbeat rate         : slow
Hash                   : l3-src-dst
Aggregate mode         : active
```

Displaying LACP aggregates:

```
switch# show lacp aggregates
Aggregate-name         : lag1
Aggregated-interfaces : 1/1/27 1/1/28 1/1/29
Heartbeat rate         : slow
Hash                   : l3-src-dst
Aggregate mode         : active
Aggregate-name         : lag2
Aggregated-interfaces : 1/1/48
Heartbeat rate         : slow
Hash                   : l2-src-dst
Aggregate mode         : passive
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Operator (`>`) or Manager (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

`show lacp configuration`

Syntax

```
show lacp configuration [vsx-peer]
```

Description

Displays global LACP configuration.

Parameter	Description
`vsx-peer`	Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Examples

Displaying global LACP configuration (output is applicable for 8400 series switches):

```
switch# show lacp configuration
System-id       : 70:72:cf:ef:fc:d9
System-priority : 65534
Hash            : l3-src-dst
```

Displaying global LACP configuration:

```
switch# show lacp configuration
System-id       : 98:f2:b3:68:40:a0
System-priority : 65534
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Operator (`>`) or Manage r (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

`show lacp interfaces`

Syntax

```
show lacp interfaces [<IFNAME>] [vsx-peer]
```

Description

Displays an LACP configuration of the physical interfaces, including VSXs. If an interface name is passed as argument, it only displays an LACP configuration of a specified interface.

Parameter	Description
`<IFNAME>`	Optional: Specifies an interface name.
`vsx-peer`	Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output f rom the VSX peer switch is not displayed. This parameter is avai lable on switches that support VSX.

Examples

This example displays an LACP configuration of the physical interfaces. One of the interfaces has the **lacp-block** forwarding state. If a VSX switch has loop protect enabled on an interface and a loop occurs, VSX blocks the interface to stop the loop. The forwarding state of the blocked interface is set to **lacp-block**.

```
switch# show lacp interfaces
State abbreviations :
A - Active         P - Passive      F - Aggregable I - Individual
S - Short-timeout L - Long-timeout N - InSync     O - OutofSync
C - Collecting    D - Distributing
X - State m/c expired              E - Default neighbor state
Actor details of all interfaces:
----------------------------------------------------------------------------
--------
Intf   Aggr    Port    Port   State   System-id         System  Aggr
Forwarding
       name    id      Pri                              Pri     Key  State
----------------------------------------------------------------------------
--------
1/1/1  lag10   17      1      ALFOE   70:72:cf:37:a3:5c  20      10   lacp-
block
1/1/2  lag128  69      1      ALFNCD  70:72:cf:37:a3:5c  20      128  up
1/1/3  lag128  14      1      ALFNCD  70:72:cf:37:a3:5c  20      128  up
1/1/4  lag128                                                         down
1/1/5  lag20                                                          up
Partner details of all interfaces:
----------------------------------------------------------------------------
--
Intf   Aggr    Partner Port     State     System-id         System   Aggr
       name    Port-id Pri                                  Priority Key
----------------------------------------------------------------------------
--
1/1/1  lag10   0       65534    PLFOEX  00:00:00:00:00:00 65534    0
1/1/2  lag128  69      1        PLFNCD  70:72:cf:8c:60:a7 65534    128
1/1/3  lag128  14      1        PLFNCD  70:72:cf:8c:60:a7 65534    128
1/1/4  lag128
1/1/5  lag20
```

Displaying static LAG:

NOTE

lacp fallback-staticcannot be configured on static lag. Attempts to configure **lacp fallback-static** on a static LAG results in the following message:

Cannot enable LACP-fallback static on static LAG.

```
switch# show lacp interfaces
State abbreviations :
A - Active        P - Passive      F - Aggregable I - Individual
S - Short-timeout L - Long-timeout N - InSync     O - OutofSync
C - Collecting    D - Distributing
X - State m/c expired              E - Default neighbor state
Actor details of all interfaces:
-----------------------------------------------------------------------------
--
Intf   Aggr   Port    Port   State   System-id         System  Aggr
Forwarding
       Name   Id      Pri                              Pri     Key  State
-----------------------------------------------------------------------------
--
1/1/1  lag10                                                        up
1/1/2  lag10                                                        up
Partner details of all interfaces:
-----------------------------------------------------------------------------
--
Intf   Aggr   Port    Port   State   System-id         System  Aggr
       Name   Id      Pri                              Pri     Key
-----------------------------------------------------------------------------
--
1/1/1  lag10
1/1/2  lag10
State abbreviations :
A - Active        P - Passive      F - Aggregable I - Individual
S - Short-timeout L - Long-timeout N - InSync     O - OutofSync
C - Collecting    D - Distributing
X - State m/c expired              E - Default neighbor state
Actor details of all interfaces:
-----------------------------------------------------------------------------
--
Intf   Aggr   Port    Port   State   System-id         System  Aggr
Forwarding
       Name   Id      Pri                              Pri     Key  State
-----------------------------------------------------------------------------
--
1/1/1  lag10                                                        up
1/1/2  lag10                                                        up
Partner details of all interfaces:
-----------------------------------------------------------------------------
--
```

```
Intf     Aggr     Port      Port     State    System-id             System   Aggr
         Name     Id        Pri                                     Pri      Key
-----------------------------------------------------------------------------
--
1/1/1   lag10
1/1/2   lag10
```

Displaying an LACP configuration of the 1/1/1 interface:

```
switch# show lacp interfaces 1/1/1
State abbreviations :
A - Active          P - Passive       F - Aggregable I - Individual
S - Short-timeout L - Long-timeout N - InSync      O - OutofSync
C - Collecting     D - Distributing
X - State m/c expired              E - Default neighbor state
Aggregate-name : lag1
-------------------------------------------------
                       Actor              Partner
-------------------------------------------------
Port-id             | 28                   | 31
Port-priority       | 1                    | 1
Key                 | 1                    | 1
State               | ALFNCD               | ALFNCD
System-id           | 98:f2:b3:68:40:a0  | 98:f2:b3:68:60:a6
System-priority     | 65534                | 65534
```

Displaying an LACP configuration after loop-protect is enabled on the primary VSX switch:

```
switch# show lacp interfaces
State abbreviations :
A - Active          P - Passive       F - Aggregable I - Individual
S - Short-timeout L - Long-timeout N - InSync      O - OutofSync
C - Collecting     D - Distributing
X - State m/c expired              E - Default neighbor state
Actor details of all interfaces:
-----------------------------------------------------------------------------
--
Intf     Aggr        Port  Port  State   System-ID         System Aggr
Forwarding
         Name        Id    Pri                             Pri     Key  State
-----------------------------------------------------------------------------
--
1/4/14  lag1(mc)    206   1     ALFNCD  f8:60:f0:06:49:00 65534   1    up
1/5/15  lag2(mc)                                                        down
Partner details of all interfaces:
-----------------------------------------------------------------------------
--
```

```
Intf     Aggr        Port  Port  State   System-ID         System Aggr
         Name        Id    Pri                             Pri    Key
-----------------------------------------------------------------------------
--
1/4/14   lag1(mc)    130   1     ALFNCD  f8:60:f0:06:87:00 65534  1
1/5/15   lag2(mc)
```

Displaying an LACP configuration after loop-protect is enabled on the secondary VSX switch:

```
switch# show lacp interfaces
State abbreviations :
A - Active        P - Passive      F - Aggregable I - Individual
S - Short-timeout L - Long-timeout N - InSync     O - OutofSync
C - Collecting    D - Distributing
X - State m/c expired              E - Default neighbor state
Actor details of all interfaces:
-----------------------------------------------------------------------------
--
Intf     Aggr        Port  Port  State   System-ID         System Aggr
Forwarding
         Name        Id    Pri                             Pri    Key  State
-----------------------------------------------------------------------------
--
1/3/2    lag1(mc)    1130  1     ALFNCD  f8:60:f0:06:49:00 65534  1    up
1/9/3    lag2(mc)                                                      down
Partner details of all interfaces:
-----------------------------------------------------------------------------
--
Intf     Aggr        Port  Port  State   System-ID         System Aggr
         Name        Id    Pri                             Pri    Key
-----------------------------------------------------------------------------
--
1/3/2    lag1(mc)    131   1     ALFNCD  f8:60:f0:06:87:00 65534  1
1/9/3    lag2(mc)
```

Displaying an LACP configuration with LACP fallback:

```
switch# show lacp interfaces
State abbreviations :
A - Active        P - Passive      F - Aggregable I - Individual
S - Short-timeout L - Long-timeout N - InSync     O - OutofSync
C - Collecting    D - Distributing
X - State m/c expired              E - Default neighbor state
Actor details of all interfaces:
-----------------------------------------------------------------------------
------
Intf        Aggr        Port  Port  State   System-ID         System Aggr
```

```
Forwarding
           Name       Id     Pri                            Pri     Key
State
----------------------------------------------------------------------------
------
1/1/4      lag10      5      1      IE      ec:eb:b8:e4:29:00 65534   10   up
1/1/5      lag10      6      1      IE      ec:eb:b8:e4:29:00 65534   10
lacp-block
1/1/6      lag10      7      1      IE      ec:eb:b8:e4:29:00 65534   10
lacp-block
1/3/27     lag10      156    1      IE      ec:eb:b8:e4:29:00 65534   10
lacp-block
1/1/9      lag20(mc)  9      1      IE      ec:eb:b8:e4:29:00 65534   10   up
Partner details of all interfaces:
----------------------------------------------------------------------------
------
Intf       Aggr       Port   Port   State   System-ID         System Aggr
           Name       Id     Pri                              Pri    Key
----------------------------------------------------------------------------
------
1/1/4      lag10      0      0      IE      00:00:00:00:00:00 0      0
1/1/5      lag10      0      0      IE      00:00:00:00:00:00 0      0
1/1/6      lag10      0      0      IE      00:00:00:00:00:00 0      0
1/3/27     lag10      0      0      IE      00:00:00:00:00:00 0      0
1/1/9      lag20(mc)  0      0      IE      00:00:00:00:00:00 0      0
```

Command History

Release	Modification
10.11	LACP fallback - static added.
10.11	LACP fallback added on VSX - supported platforms.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Operator (`>`) or Manager (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

`show lag`

Syntax

```
show  lag <LAG-ID>
```

Description

Displays the lag.

Parameter	Description
<LAG-ID>	Specifies the lag ID.

Examples

Displaying the lag.

```
switch# show lag
System-ID       : f4:03:43:80:4a:00
System-priority : 65534
Hash            : l3-src-dst
Aggregate lag1 is down
   Admin state is down
   Description :
   Type                          : normal
   Lacp Fallback                 : n/a
   MAC Address                   : f4:03:43:80:4a:00
   Aggregated-interfaces         :
   Aggregation-key               : 1
   Aggregate mode                : static
   LACP rate                     : n/a
   Speed                         : 0 Mb/s
   Mode                          : routed
Aggregate lag128 is down
  Admin state is down
  Description :
  Type                          : normal
  Lacp Fallback                 : n/a
```

```
  MAC Address                    : f4:03:43:80:4a:00
-- MORE --, next page: Space, next line: Enter, quit: q
```

Displaying the lag when `lacp fallback-static` is enabled.

```
switch# show lag
System-ID       : 90:20:c2:24:60:00
System-priority : 65534
Aggregate lag1 is up
   Admin state is up
   Description :
   Type                            : normal
   Lacp Fallback                   : Enabled
   MAC Address                     : 90:20:c2:24:60:00
   Aggregated-interfaces           : 1/1/1 1/1/2 1/1/3 1/1/46 1/1/47 1/1/48
   Aggregation-key                 : 1
   Aggregate mode                  : active
   Hash                            : l3-src-dst
   LACP rate                       : slow
   Speed                           : 1000 Mb/s
   Mode                            : trunk
```

Release	Modification
10.11	LACP fallback - static added.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Operator (`>`) or Manage r (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

`show running-config interface`

Syntax

```
show running-config interface
   [IFNAME | IFRANGE | lag <LAGNUM.ID> | loopback <NUMBER> | mgmt | persona
| tunnel <ID> | vlan <ID> | vsx-peer | vxlan] {vsx-peer}
```

Description

Displays the running configuration for the interface.

Parameter	Description
`<IFNAME>`	Displays the interface name.
`<IFRANGE>`	Displays the PORT identifier range.
`lag <LAGNUM.ID>`	Displays the LAG interface information. Range: 1 to 4094.
`loopback <NUMBER>`	Displays the loopback interface information. Range: 0 to 255.
`mgmt`	Displays the management interface information.
`persona`	Displays the interface persona information.
`tunnel <ID>`	Displays the tunnel interface information. Range: 1 to 1255. • ◦ **gre ip**: Displays the details of GRE IPv4 tunnel interface . • ◦ **ip 6in4**: Displays the details of IPv6 in IPv4 tunnel infor mation. • ◦ **ip 6in6**: Displays the details of IPv6 in IPv6 tunnel infor mation.
`vlan <ID>`	Displays the VLAN interface information. Range: 1 to 4094.

Parameter	Description
`vsx-peer`	Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output f rom the VSX peer switch is not displayed. This parameter is avai lable on switches that support VSX.
`vxlan`	Displays the VXLAN interface information. Default: 1.

Examples

Displaying the running configuration for the interface.

```
switch# show running-config interface
interface lag 10
    no shutdown
    mtu 9000
    description "Server uplink LAG"
    persona custom server-uplink attach
interface lag 11
    no shutdown
    mtu 9000
    description "Server uplink LAG"
    persona custom server-uplink
interface persona lag server-uplink
    no shutdown
    mtu 9000
    description "Server uplink LAG"
```

Displaying the running configuration for interface lag.

```
switch# show running-config interface lag
interface lag 10 multi-chassis
   no shutdown
   no routing
   vlan trunk native 1
   vlan trunk allowed 10-12
   lacp mode active
   exit
interface lag 11 multi-chassis
   no shutdown
   no routing
   vlan trunk native 1
   vlan trunk allowed 10-12,2001
   lacp mode active
   exit
interface lag 256
```

```
   description VSX_ISL
   no shutdown
   no routing
   vlan trunk native 1 tag
   vlan trunk allowed all
   lacp mode active
   exit
```

Displaying the running configuration for interface lag with **lacp fallback-static** configured.

```
switch# show running-config interface lag
interface lag 1
   no shutdown
   no routing
   vlan trunk native 1
   vlan trunk allowed all
   lacp mode active
   lacp fallback-static
```

Displaying the running configuration for interface persona.

```
switch# show running-config interface persona
interface persona access
    shutdown
    no routing
    vlan access 1
    exit
interface persona persona1
    shutdown
    no routing
    vlan access 1
    exit
interface persona persona2
    shutdown
    no routing
    vlan access 1
    exit
```

Command History

Release	Modification
10.17	Added support for persona LAG interfaces.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

shutdown

Syntax

```
shutdown
no shutdown
```

Description

Sets every interface in the LAG operationally down.

The **no** form of this command sets every interface operationally up.

Examples

Setting every interface in the LAG to shutdown:

```
switch(config)# interface lag 1
switch(config-lag-if)#
```

Resetting every interface in the LAG to the default (up):

```
switch(config)# interface lag 1
switch(config-lag-if)# no shutdown
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	`config-lag-if`	Administrators or local user group members with execution righ ts for this command.

`vlan trunk native`

Syntax

```
vlan trunk native <VLAN-ID>
no vlan trunk native [<VLAN-ID>]
```

Description

Assigns a native VLAN ID to a LAG interface.

The **no** form of this command removes a native VLAN from a LAG interface and assigns VLAN ID 1 as its native VLAN.

Parameter	Description
<VLAN-ID>	Specifies the number of the VLAN ID to assign. The VLAN ID m ust exist. Maximum number of VLANs supported: 512 (4100i) Maximum number of VLANs supported: 512 (6000 and 6100) Maximum number of VLANs supported: 2048 (5420, 6200) Maximum number of VLANs supported: 4096 (6300, 6400) Maximum number of VLANs supported: 4096 (8320) Maximum number of VLANs supported: 4096 (8325) Maximum number of VLANs supported: 4096 (10000) Maximum number of VLANs supported: 4096 (9300/9300S) Maximum number of VLANs supported: 4096 (8360) Maximum number of VLANs supported: 1024 (8100) Maximum number of VLANs supported: 4096 (8400) VLAN ID range: 2 to 4094.

Usage

By default, VLAN ID 1 is assigned as the LAG VLAN ID for all LAG interfaces. VLANs can only be assigned to a nonrouted (layer 2) interface or LAG interface.

Only one VLAN ID can be assigned as the native VLAN. For the interface to forward the native VLAN traffic, the interface has to be allowed explicitly by entering **vlan trunk allowed <ID>** where the ID is the native VLAN ID. This setting is also applicable to the physical interface.

Examples

Configuring a layer 2 dynamic aggregation group with native VLAN ID **1** assigned to LAG **1**:

For HPE Aruba Networking 5420, 6000, 6100, and 6200 Switch series:

```
switch(config)# interface lag 1
switch(config-lag-if)# no shutdown
switch(config-lag-if)# lacp mode active
switch(config-lag-if)# vlan trunk native 1
switch(config-lag-if)# vlan trunk allowed 1
```

For HPE Aruba Networking 6300, 6400, 8100, 8320, 8325, 8360, 8400, 9300/9300S, and 10000 Switch series:

```
switch(config)# interface lag 1
switch(config-lag-if)# no shutdown
switch(config-lag-if)# no routing
switch(config-lag-if)# lacp mode active
switch(config-lag-if)# vlan trunk native 1
switch(config-lag-if)# vlan trunk allowed 1
```

Configuring a layer 2 dynamic aggregation group with native VLAN ID **20** assigned to LAG **1**:

For HPE Aruba Networking 5420, 6000, 6100, and 6200 Switch series:

```
switch(config)# interface lag 1
switch(config-lag-if)# no shutdown
switch(config-lag-if)# lacp mode active
switch(config-lag-if)# vlan trunk native 20
switch(config-lag-if)# vlan trunk allowed 20
```

For HPE Aruba Networking 6300, 6400, 8100, 8320, 8325, 8360, 8400, 9300/9300S, and 10000 Switch series:

```
switch(config)# interface lag 1
switch(config-lag-if)# no shutdown
switch(config-lag-if)# no routing
switch(config-lag-if)# lacp mode active
switch(config-lag-if)# vlan trunk native 20
switch(config-lag-if)# vlan trunk allowed 20
```

Removing a native VLAN from LAG 1:

```
switch(config)# interface lag 1
switch(config-lag-if)# no vlan trunk native
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	`config-if` `config-lag-if`	Administrators or local user group members with execution righ ts for this command.

Smartlink

Smartlink provides simple and fast-converging link redundancy in network topologies with dual uplink between different layers of the network. It requires an active (primary) and backup (secondary) link. The active link carries the traffic. If the active link fails, a switchover is triggered and the traffic is directed to the backup link.

The active interface forwards traffic for a group of VLANs (referred to as protected VLAN group). The secondary interface is in backup mode for this protected group. If the active port goes down, the backup port starts forwarding traffic for the protected VLAN group. If the active port recovers, it switches to backup mode and does not forward traffic. Secondary port continues forwarding traffic.

If preemption is enabled, a failed active port (that has recovered) becomes active after the configured "preemption-delay" time has elapsed.

Smartlink is supported by the following switch platforms:

- 4100 Switch Series
- 5420 Switch Series
- 6000 Switch Series
- 6100 Switch Series

- 6200 Switch Series
- 6300 Switch Series
- 6400 Switch Series
- 8100 Switch Series
- 8360 Switch Series

Benefits

Smartlink provides faster failover compared to STP. If an active link fails, a Smartlink group contains configuration information that determines which port should be forwarding for a protected VLAN group.

Subtopics

Guidelines and limitations
Smartlink commands

Guidelines and limitations

- For faster convergence of routed traffic over Smartlink ports, ip neighbor-flood must be enabled on respective SVI interfaces.
- Smartlink uses ERPS copp class for flush packets.
- The Aruba 6000 and 6100 Switch Series use separate Smartlink coPP policies as ERPS is not supported in matrix platforms.

Limitations:

- VSX, ISL and MCLAGs cannot be included in Smartlink groups.
- Switches with both Smartlink and STP enabled exclude Smartlink ports from STP.
- On switches with both Smartlink and STP enabled, loops involving Smartlink and STP are not detected.
- On switches with both Smartlink and ERPS enabled, loops involving Smartlink and ERPS are not detected.
- ERPS and Smartlink cannot be enabled on the same port.
- Dynamic VLANs (MVRP) and Smartlink cannot be enabled on the same port.
- Loop Protect and Smartlink cannot be enabled on the same port.
- Multicast fast convergence is not supported.

- Uplink Failure Detection (UFD) is not supported.
- MIB and WebUI are not supported.

NOTE

- VLANs that include Smartlink ports must be included in the protected VLAN list of at least one Smartlink group. If a VLAN includes Smartlink ports and is not included in the protected VLAN list, the VLAN-port combination will not be managed by Smartlink or STP, resulting in an undefined port state for the VLAN, which will cause a loop in the network.
- When using UDLD with Smartlinks, redundancy switchover is not hitless and will result in traffic loss.

Smartlink commands

Select a command from the list in the left navigation menu.

Subtopics

Configuration commands
Group context commands
Display commands
Supportability commands

Configuration commands

Select a command category from the list in the left navigation menu.

Subtopics

smartlink group
smartlink recv-control-vlan

`smartlink group`

Syntax

```
smartlink group <GROUP-ID>
no smartlink group <GROUP-ID>
```

Description

Creates a Smartlink group with specified ID.

The **no** form of this command removes the Smartlink group and all associated configurations for a specified ID.

Parameter	Description
`<GROUP-ID>`	Specifies ID for the Smartlink group.

Usage

The maximum number of Smartlink groups is 24.

Examples

Configuring a Smartlink group:

```
switch(config)# smartlink group 2
switch(config-smartlink-2)#
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420 6000 6100 6200	`config`	Administrators or local user group members with execution righ ts for this command.

Platforms	Command context	Authority
6300 6400 8100 8360		

smartlink recv-control-vlan

Syntax

```
smartlink recv-control-vlan <VID-LIST>
no smartlink recv-control-vlan <VID-LIST>
```

Description

Configures control VLANs to receive flush messages.

The **no** form of this command disables VLANs from receiving flush messages.

Parameter	Description
<VID-LIST>	Specifies VLAN ID.

Usage

- Configure this command on uplink devices where MAC flush is required.
- A flush message clears stale MAC and ARP entries enabling fast traffic convergence.

Examples

Configuring control VLAN to receive flush messages:

```
switch(config)# smartlink recv-control-vlan 2,3
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6000 6100 6200 6300 6400 8100 8360	`config`	Administrators or local user group members with execution righ ts for this command.

Group context commands

Select a command category from the list in the left navigation menu.

Subtopics

- **description**
- **diag-dump smartlink basic**
- **primary-port**
- **smartlink group secondary-port**
- **control-vlan**
- **protected-vlans**
- **preemption**
- **preemption-delay**

`description`

Syntax

```
description <DESC>
no description
```

Description

Adds description to a Smartlink group.

The **no** form of this command removes a description from a Smartlink group.

Parameter	Description
`<DESC>`	Specifies description for a Smartlink group. 1 to 64 printable AS CII characters are allowed.

Examples

Adding a description to a Smartlink group:

```
switch(config)# smartlink group 3
switch(config-smartlink-3)# Description for group 3
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6000 6100 6200 6300	`config-smartlink-<GROUP>`	Administrators or local user group members with execution righ ts for this command.

Platforms	Command context	Authority
6400		
8100		
8360		

`diag-dump smartlink basic`

Syntax

```
diag-dump smartlink basic
```

Description

Dumps the Smartlink configuration, state and statistics.

Examples

Dump of Smartlink configuration, state, and statistics:

```
switch# diag-dump smartlink basic
===========================================================================
[Start] Feature smartlink Time : Tue Jul  7 10:08:31 2020
===========================================================================
---------------------------------------------------------------------------
[Start] Daemon smartlinkd
---------------------------------------------------------------------------
SL Group 1: Primary port 1/1/1 Secondary port 1/1/2 Control VLAN 4,
            Preemption disabled, Preemption-delay 1 Preemption Timer OFF,
            State primary_with_backup, Active port PRIMARY, Backup port
SECONDARY
Port 1/1/1: member_groups 1 SL Groups ids: 1, 0
Port 1/1/2: member_groups 1 SL Groups ids: 1, 0
or
SL Group 1: Primary port lag1   (mclag: local_up_remote_up)
            Secondary port lag2 (mclag: local_down_remote_up), Control VLAN
4,
            Preemption disabled, Preemption-delay 1 Preemption Timer OFF,
            State primary_with_backup, Active port PRIMARY, Backup port
SECONDARY
Port lag1: member_groups 1 SL Groups ids: 1, 0
Port lag2: member_groups 1 SL Groups ids: 1, 0
```

```
VSX Oper Status: Primary/Secondary/NA
--------------------------------------------------------------------------
[End] Daemon smartlinkd
--------------------------------------------------------------------------
--------------------------------------------------------------------------
[Start] Daemon ops-switchd
--------------------------------------------------------------------------
Group-ID | Port Name | Port Status | Vlan-ID | HW-Port-State | Vlan-Type
1        | 1/1/1     | Active      | 4       | Forwarding    | Control
1        | 1/1/1     | Active      | 3       | Forwarding    | Protected
1        | 1/1/1     | Active      | 2       | Forwarding    | Protected
1        | 1/1/1     | Active      | 1       | Forwarding    | Protected
1        | 1/1/2     | Backup      | 4       | Blocking      | Control
1        | 1/1/2     | Backup      | 3       | Blocking      | Protected
1        | 1/1/2     | Backup      | 2       | Blocking      | Protected
1        | 1/1/2     | Backup      | 1       | Blocking      | Protected
--------------------------------------------------------------------------
[End] Daemon ops-switchd
--------------------------------------------------------------------------
==========================================================================
[End] Feature smartlink
==========================================================================
Diagnostic-dump captured for feature smartlink
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6000 6100 6200 6300 6400	Operator (>) or Manage r (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

Platforms	Command context	Authority
8100 8360		

primary-port

Syntax

```
primary-port <INTERFACE-NAME>
no primary-port
```

Description

Configures primary port for a Smartlink group.

The **no** form of this command removes primary port from a Smartlink group.

Parameter	Description
`<INTERFACE-NAME>`	Specifies interface for primary port.

Examples

Configuring primary port for a Smartlink group:

```
switch(config)# smartlink group 3
switch(config-smartlink-3)# primary-port 1/1/1
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420 6000 6100 6200 6300 6400 8100 8360	`config-smartlink-<GROUP>`	Administrators or local user group members with execution righ ts for this command.

smartlink group secondary-port

Syntax

```
secondary-port <INTERFACE-NAME>
no secondary-port
```

Description

Configures secondary port for a Smartlink group.

The **no** form of this command removes secondary port from a Smartlink group.

Parameter	Description
`<INTERFACE-NAME>`	Specifies interface for secondary port.

Examples

Configuring secondary port for a Smartlink group:

```
switch(config)# smartlink group 3
switch(config-smartlink-3)# secondary-port 1/1/2
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6000 6100 6200 6300 6400 8100 8360	`config-smartlink-<GROUP>`	Administrators or local user group members with execution righ ts for this command.

control-vlan

Syntax

```
control-vlan <VLAN-ID>
no control-vlan <VLAN-ID>
```

Description

Configures control VLAN in a Smartlink group.

The **no** form of this command removes control VLAN from a Smartlink group.

Parameter	Description
	Specifies VLAN ID for a Smartlink group.

Parameter	Description
<VLAN-ID>	

Usage

- In a Smartlink group, the control VLAN is used to send flush messages.
- Control VLAN is configured on the device intended to send flush messages.
- Each Smartlink group must use a unique control VLAN.
- Control VLAN is protected in the Smartlink group to avoid loops.

Examples

Configuring control VLAN in a Smartlink group:

```
switch(config)# smartlink group 3
switch(config-smartlink-3)# control-vlan 10
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6000 6100 6200 6300 6400 8100 8360	`config-smartlink-<GROUP>`	Administrators or local user group members with execution righ ts for this command.

`protected-vlans`

Syntax

```
protected-vlans <VLAN-ID-LIST>
no protected-vlans <VLAN-ID-LIST>
```

Description

Specifies VLANs protected by a Smartlink group.

The **no** form of this command removes VLANs protected by a Smartlink group.

Parameter	Description
`<VLAN-ID-LIST>`	Specifies list of VLAN IDs. Range is 1 to 4094.

Examples

Configuring protected VLANs for a Smartlink group.:

```
switch(config)# smartlink group 3
switch(config-smartlink-3)# protected-vlans 1, 10-50
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6000	`config-smartlink-<GROUP>`	Administrators or local user group members with execution righ ts for this command.

Platforms	Command context	Authority
6100 6200 6300 6400 8100 8360		

`preemption`

Syntax

```
preemption
no preemption
```

Description

Configures preemption in a Smartlink group.

The **no** form of this command disables preemption in a Smartlink group.

Usage

- If preemption is enabled, a recovered primary port preempts the active interface after the configured preemption delay.
- If preemption is disabled, a recovered primary port serves as a backup interface and does not forward traffic.

Examples

Configuring preemption in a Smartlink group:

```
switch(config)# smartlink group 3
switch(config-smartlink-3)# preemption
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420 6000 6100 6200 6300 6400 8100 8360	`config-smartlink-<GROUP>`	Administrators or local user group members with execution rights for this command.

`preemption-delay`

Syntax

```
preemption-delay <SECONDS>
no preemption-delay
```

Description

Specifies preemption delay for a Smartlink group.

The **no** form of this command removes previously configured preemption delay from a Smartlink group and sets it to the default of 1 second.

Parameter	Description
	Specifies preemption delay in seconds. Range is 0 to 300 seconds.

Parameter	Description
<SECONDS>	

Usage

When preemption is enabled, a recovered primary port always preempts the active interface after the configured preemption delay.

Examples

Configuring preemption delay on a Smartlink group:

```
switch(config)# smartlink group 3
switch(config-smartlink-3)# preemption
switch(config-smartlink-3)# preemption-delay 10
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6000 6100 6200 6300 6400 8100 8360	config-smartlink-*<GROUP>*	Administrators or local user group members with execution righ ts for this command.

Display commands

Select a command category from the list in the left navigation menu.

Subtopics

- show smartlink group
- show smartlink group all
- show smartlink group detail
- show smartlink flush-statistics
- clear smartlink group statistics
- clear smartlink flush-statistics
- show running-config

`show smartlink group`

Syntax

```
show smartlink group <GROUP-ID>
```

Description

Shows information for a specific Smartlink group.

Parameter	Description
`<GROUP-ID>`	Specifies Smartlink group ID.

Examples

Showing Smartlink group information:

```
switch# show smartlink group 1
Smartlink Group 1 Information:
==============================
Group description          : Uplink1
Protected VLANs            : 20-30
Control VLAN               : 10
Preemption                 : ON
Preemption Delay           : 10
```

```
Ports  Role      State      Flush Count Last Flush Time
------ --------- ---------- ----------- -------------------------
1/1/1  Primary   Active     2           Sat Oct 17 19:09:10 2020
1/1/2  Secondary Backup     0
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6000 6100 6200 6300 6400 8100 8360	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show smartlink group all

Syntax

```
show smartlink group all
```

Description

Shows information for all configured Smartlink groups.

Examples

Showing information for all configured Smartlink groups:

```
switch# show smartlink group all
Smartlink Group Information:
==============================
     Primary Secondary  Active   Backup   Ctrl      Preemption Preemption
Grp  Port    Port       Port     Port     Vlan                 Delay
---- ------- ---------  ------   -------  --------- ---------- ----------
1    1/1/1   1/1/2      1/1/1    1/1/2    10           OFF         1
2    1/1/5   1/1/6      1/1/5    1/1/6    11           OFF         1
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6000 6100 6200 6300 6400 8100 8360	Operator (`>`) or Manage r (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

`show smartlink group detail`

Syntax

```
show smartlink group detail
```

Description

Shows detailed information for all configured Smartlink groups.

Examples

Showing detailed information for all configured Smartlink groups:

```
switch# show smartlink group detail
Smartlink Group 1 Information:
===============================
Protected VLAN                : 1-3
Control VLAN                  : 1
Preemption                    : OFF
Preemption Delay              : 1
Ports    Role         State        Flush Count  Last Flush Time
-------- ------------ ------------ ------------ ------------------------
1/3/1    Primary      Backup       0
1/3/2    Secondary    Active       0
Smartlink Group 2 Information:
===============================
Protected VLAN                : 4-6
Control VLAN                  : 4
Preemption                    : OFF
Preemption Delay              : 1
Ports    Role         State        Flush Count  Last Flush Time
-------- ------------ ------------ ------------ ------------------------
1/3/2    Primary      Active       0
1/3/1    Secondary    Backup       0
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6000 6100 6200 6300 6400	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Platforms	Command context	Authority
8100 8360		

show smartlink flush-statistics

Syntax

```
show smartlink flush-statistics
```

Description

Shows information for received flush messages.

Usage

This command must be executed on an uplink or peer device configured with **recv-control-vlan**.

Examples

Showing information for received flush messages:

```
switch# show smartlink flush-statistics
Last Flush Packet Detail:
========================
Flush Packets Received                     : 2
Last Flush Packet Received On Interface    : 1/1/1
Last Flush Packet Received On              : Sat Oct 17 19:09:10 2020
Device Id Of Last Flush Packet Received    : 5065f3-127080
Control VLAN Of Last Flush Packet Received : 10
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6000 6100 6200 6300 6400 8100 8360	Operator (`>`) or Manager (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

`clear smartlink group statistics`

Syntax

```
clear smartlink group [<GROUP-ID>] statistics
```

Description

Clears Smartlink statistics for the specified Smartlink group or all Smartlink groups.

Parameter	Description
`<GROUP-ID>`	Specifies Smartlink group.

Examples

Clearing Smartlink statistics for a specified Smartlink group:

```
switch# clear smartlink group 1 statistics
```

Clearing all Smartlink statistics for all Smartlink groups:

```
switch(config)# clear smartlink group statistics
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420 6000 6100 6200 6300 6400 8100 8360	Operator (`>`) or Manager (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

`clear smartlink flush-statistics`

Syntax

```
clear smartlink flush-statistics
```

Description

Clears Smartlink flush statistics.

Usage

This command must be executed on the uplink device configured with **recv-control-vlan**.

Examples

Clearing Smartlink flush statistics:

```
switch# clear smartlink flush-statistics
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420 6000 6100 6200 6300 6400 8100 8360	Operator (`>`) or Manager (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

`show running-config`

Syntax

```
show running-config
```

Description

Shows current running configuration.

Examples

Showing currently running configuration:

```
switch# configure terminal
switch(config)# smartlink group 1
switch(config-smartlink-1)# description Uplink1
switch(config-smartlink-1)# primary-port 1/1/1
switch(config-smartlink-1)# secondary-port 1/1/2
switch(config-smartlink-1)# control-vlan 10
```

```
switch(config-smartlink-1)# protected-vlans 20-30
switch(config-smartlink-1)# preemption
switch(config-smartlink-1)# preemption-delay 10
switch(config)# smartlink group 2
switch(config-smartlink-2)# primary-port 1/1/8
switch(config-smartlink-2)# secondary-port 1/1/9
switch(config-smartlink-2)# control-vlan 11
switch(config-smartlink-2)# protected-vlans 20-30
switch# show running-config
Current configuration:
!
!
!
smart-link group 1
  primary-port 1/1/1
  secondary-port 1/1/2
  control-vlan 10
  protected-vlans 20-30
  preemption
  preemption-delay 10
  exit
smart-link group 2
  primary-port 1/1/8
  secondary-port 1/1/9
  control-vlan 11
  protected-vlans 20-30
  exit
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6000 6100	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Platforms	Command context	Authority
6200		
6300		
6400		
8100		
8360		

Supportability commands

Select a command category from the list in the left navigation menu.

Subtopics

`show capacities smartlink`

Syntax

```
show capacities smartlink | show capacities-status smartlink
```

Description

Shows Smartlink capacities or Smartlink capacities and status.

Examples

Showing Smartlink capacities:

```
switch# show capacities smartlink
System Capacities: Filter SMARTLINK
Capacities
Name                                                                        Value
-----------------------------------------------------------------------------
--------
Maximum number of SMARTLINK GROUPS configurable in a
system                             24
```

Showing Smartlink capacities and status:

```
switch# show capacities-status smartlink
System Capacities Status: Filter SMARTLINK
Capacities Status Name                                              Value
Maximum
------------------------------------------------------------------------------
--------
Number of SMARTLINK GROUPS currently configured
1      24
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
4100i 5420 6000 6100 6200 6300 6400 8100 8360	Operator (`>`) or Manage r (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

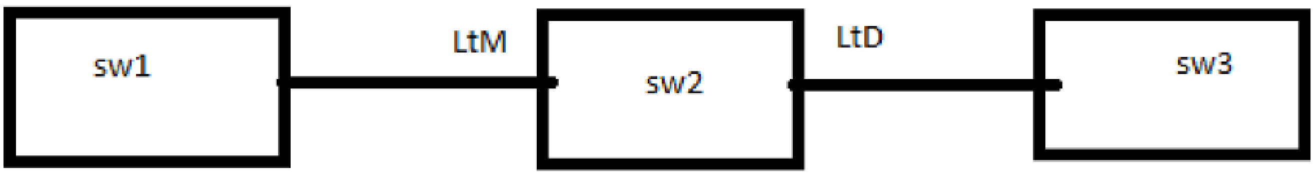
UFD (Uplink Failure Detection)

Uplink Failure Detection (UFD) is used to help achieve network path redundancy. UFD monitors (tracks the forwarding state of) the interfaces/LAGs configured as Links-to-Monitor (LtM) and when **all LtM links go down**, UFD disables the interfaces/LAGs configured as Links-to-Disable (LtD). If **any of the LtM links come back up**, then all the LtD links are brought back up.

This process triggers the re-convergence of the traffic to the redundant path that is typically set up using another switch or network. A common example is the teaming NIC software in servers that is used to fail over from the primary NIC to the secondary NIC upon primary NIC failure.

To avoid unnecessary switching in the downlink redundant path during a frequent network flap in the uplink ports, delays can be configured. For example, if all the monitored uplinks are still down after the configured **down** delay, all the links to disable interfaces/LAGs are brought down. Similarly, if any of the monitored uplinks are still up after the configured **up** delay, all the disabled interfaces/LAGs are brought back up.

In this simplistic topology, switch sw2 uses UFD to monitor the links (LtM) to switch sw1, disabling the links (LtD) to switch sw3 upon failure of the links from switch sw2 to switch sw1. When sw3 detects that the links from switch sw2 have gone down, it then switches to its redundant path.



> **NOTE**
> Although UFD can be used alone, consider using it with Smartlink which automates fail over from links that have gone down to redundant links. Smartlink is available on the 5420, 6200, 6300, 6400, 8100, and 8360 Switch Series.

Guidelines and limitations

- UFD is supported only on L2 interfaces and LAGs. It is not supported on ROP and SVI.
- UFD is not supported with VSX, meaning that ISL and MCLAGs are not supported.

Subtopics

Basic UFD configuration
UFD (Uplink Failure Detection) commands

Basic UFD configuration

To help understand how to configure UFD, a basic configuration is presented, followed by detailed descriptions of the available commands under UFD (Uplink Failure Detection) commands.

Enabling UFD:

```
switch(config)# ufd enable
```

Creating UFD session 1 and then entering its context:

```
switch(config)# ufd session-id 1
switch(config-ufd-1)#
```

Configuring two links to be monitored and two links to disable:

```
switch(config-ufd-1)# links-to-monitor 1/1/1,1/1/2
switch(config-ufd-1)# links-to-disable 1/1/11,1/1/12
```

Setting the up and down delays to 10 seconds:

```
switch(config-ufd-1)# delay down 10
switch(config-ufd-1)# delay up 10
```

Showing information for UFD session 1:

```
switch# show ufd session 1
UFD session-id                    : 1
UFD Links-to-Monitor status       : Up
Up Delay                          : 10 sec
Down Delay                        : 10 sec
Links-to-Monitor                  : 1/1/1,1/1/2
Links-to-Disable                  : 1/1/11,1/1/12
Last Links-to-Monitor Down Time   : 2021-11-03 15:22:05:37
```

UFD (Uplink Failure Detection) commands

Select a command from the list in the left navigation menu.

Subtopics

- debug ufd all
- delay
- links-to-disable
- links-to-monitor
- show capacities ufd
- show running-config ufd
- show-tech ufd
- show ufd
- ufd enable
- ufd session-id

`debug ufd all`

Syntax

```
debug ufd all
no debug ufd all
```

Description

Enables the UFD debug logs.

The no form of this command disables the UFD debug logs.

Examples

Enabling UFD debug logs:

```
switch(config)# debug ufd all
```

Disabling UFD debug logs:

```
switch(config)# no debug ufd all
```

Command History

Release	Modification
10.09	Command introduced.

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution righ ts for this command.

`delay`

Syntax

```
delay {down | up} <DELAY>
no delay {down | up} <DELAY>
```

Description

Within the selected UFD (Uplink Failure Detection) session context, specifies the amount of time (in seconds) to delay before bringing up or down the configured Links to Disable (LtD) after the corresponding Links to Monitor (LtM) come back up or go down.

For example, with **delay down 10**, when **all LtM links go down** and remain down after 10 seconds, UFD disables the interfaces/LAGs configured as Links-to-Disable (LtD). Similarly, with **delay up 10**, If **any of the LtM links come back up** and remain up after 10 seconds, then all the LtD links are brought back up.

> **NOTE**
> In addition to any configured delay there is an additional delay of 3 to 5 seconds before bringing any Links-to-Disable (LtD) down or back up. So with the default delay of 0 seconds, a delay of 3 to 5 seconds does occur.

The no form of this command restores the delay to its default of 0 seconds.

Parameter	Description
<DELAY>	Species the delay in seconds. Range 0 to 180 seconds. Default: 0 seconds.

Examples

Setting the up and down delays to 10 seconds:

```
switch(config)# ufd enable
switch(config)# ufd session-id 1
switch(config-ufd-1)# links-to-monitor 1/1/1,1/1/2
switch(config-ufd-1)# links-to-disable 1/1/11,1/1/12
switch(config-ufd-1)# delay down 10
switch(config-ufd-1)# delay up 10
switch(config-ufd-1)# exit
switch(config)#
```

Resetting the up and down delays to their default of 0:

```
switch(config-ufd-1)# no delay down 10
```

```
switch(config-ufd-1)# no delay up 10
```

Command History

Release	Modification
10.09	Command introduced.

Command Information

Platforms	Command context	Authority
All platforms	`config-ufd-<ID>`	Administrators or local user group members with execution righ ts for this command.

`links-to-disable`

Syntax

```
links-to-disable <IF/LAG-LIST>
no links-to-disable <IF/LAG-LIST>
```

Description

Within the selected UFD (Uplink Failure Detection) session context, specifies the interfaces or LAGs to disable when the monitored uplink interfaces go down.

For proper UFD operation, **links-to-disable** and **links-to-monitor** must both be configured. Use command **links-to-monitor** to specify a corresponding list of interfaces/LAGs to monitor.

The no form of this command deletes the specified links to disable list within the selected UFD session context.

> **NOTE**
> A LAG member interface cannot be added as a link to disable. A interface configured as a link to disable cannot be added as a LAG member interface.

Parameter	Description
`<IF/LAG-LIST>`	List of L2 interfaces or LAGs. Separate interfaces/LAGs with commas (for individual interfaces/LAGs) or hyphens (for a consecutive range of interfaces/LAGs).

Examples

Configuring two links to be disabled:

```
switch(config)# ufd enable
switch(config)# ufd session-id 1
switch(config-ufd-1)# links-to-monitor 1/1/1,1/1/2
switch(config-ufd-1)# links-to-disable 1/1/11,1/1/12
switch(config-ufd-1)# delay down 10
switch(config-ufd-1)# delay up 10
switch(config-ufd-1)# exit
switch(config)#
```

Configuring a range of interfaces to disable:

```
switch(config)# ufd session-id 2
switch(config-ufd-2)# links-to-monitor lag18-lag20
switch(config-ufd-2)# links-to-disable 1/1/3-1/1/5
switch(config-ufd-2)# exit
switch(config)#
```

Deleting the links to disable for two interfaces:

```
switch(config-ufd-1)# no links-to-disable 1/1/11,1/1/12
```

Command History

Release	Modification
10.09	Command introduced.

Command Information

Platforms	Command context	Authority
All platforms	`config-ufd-<ID>`	Administrators or local user group members with execution rights for this command.

`links-to-monitor`

Syntax

```
links-to-monitor <IF/LAG-LIST>
no links-to-monitor <IF/LAG-LIST>
```

Description

Within the selected UFD (Uplink Failure Detection) session context, specifies the uplink interfaces or LAGs to monitor for UFD.

For proper UFD operation, **links-to-monitor** and **links-to-disable** must both be configured. Use command **links-to-disable** to specify a corresponding list of interfaces/LAGs to disable if the monitored uplinks go down.

The no form of this command deletes the specified links to monitor list within the selected UFD session context.

> **NOTE**
> A LAG member interface cannot be added as a link to monitor. A interface configured as a link to monitor cannot be added as a LAG member interface.

Parameter	Description
`<IF/LAG-LIST>`	List of L2 interfaces or LAGs. Separate interfaces/LAGs with co mmas (for individual interfaces/LAGs) or hyphens (for a consec utive range of interfaces/LAGs).

Examples

Configuring two uplinks to monitor for UFD session 1:

```
switch(config)# ufd enable
switch(config)# ufd session-id 1
switch(config-ufd-1)# links-to-monitor 1/1/1,1/1/2
switch(config-ufd-1)# links-to-disable 1/1/11,1/1/12
switch(config-ufd-1)# delay down 10
switch(config-ufd-1)# delay up 10
switch(config-ufd-1)# exit
switch(config)#
```

Configuring a range of uplink LAGs to monitor for UFD session 2:

```
switch(config)# ufd session-id 2
switch(config-ufd-2)# links-to-monitor lag18-lag20
switch(config-ufd-2)# links-to-disable 1/1/3-1/1/5
switch(config-ufd-2)# exit
switch(config)#
```

Deleting both links to monitor for UFD session 1:

```
switch(config-ufd-1)# no links-to-monitor 1/1/1,1/1/2
```

Command History

Release	Modification
10.09	Command introduced.

Command Information

Platforms	Command context	Authority
All platforms	`config-ufd-<ID>`	Administrators or local user group members with execution righ ts for this command.

show capacities ufd

Syntax

```
show capacities ufd
show capacities-status ufd
```

Description

Command **show capacities ufd** shows UFD session capacity. Command **show capacities-status ufd** shows UFD session capacity and the number of UFD sessions configured.

Example

Showing UFD session capacity:

```
switch# show capacities ufd
System Capacities: Filter UFD
Capacities
```

```
Name                                                                   Value
----------------------------------------------------------------------------
---------
Maximum number of Uplink Failure Detection sessions configurable in a
system  128
```

Showing UFD session capacity and the number of UFD sessions configured:

```
switch(config)# show capacities-status ufd
System Capacities Status: Filter UFD
Capacities Status Name
Value Maximum
----------------------------------------------------------------------------
---------
Number of Uplink Failure Detection sessions currently configured
1    128
```

Command History

Release	Modification
10.09	Command introduced.

Command Information

Platforms	Command context	Authority
All platforms	Operator (`>`) or Manager (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

`show running-config ufd`

Syntax

```
show running-config ufd
```

Description

Shows the running configuration for UFD.

Example

Showing the UFD portion of running configuration information:

```
switch(config)# ufd enable
switch(config)# ufd session-id 1
switch(config-ufd-1)# links-to-monitor 1/1/1,1/1/2
switch(config-ufd-1)# links-to-disable 1/1/11,1/1/12
switch(config-ufd-1)# delay down 10
switch(config-ufd-1)# delay up 10
switch(config-ufd-1)# exit
switch(config)#
switch# show running-config ufd
Current configuration:
ufd enable
ufd session-id 1
    delay up 10
    delay down 10
    links-to-monitor 1/1/1,1/1/2
    links-to-disable 1/1/11,1/1/12
```

Command History

Release	Modification
10.09	Command introduced.

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manage r (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

`show-tech ufd`

Syntax

```
show-tech ufd
```

Description

Executes the **show ufd** command followed by the **show running-config ufd** command.

Example

Running the **show ufd** command followed by the **show running-config ufd** command:

```
switch# show tech ufd
====================================================
Show Tech executed on Tue Nov 23 11:32:08 2021
====================================================
====================================================
[Begin] Feature ufd
====================================================
*********************************
Command : show ufd
*********************************
Global UFD Status : Enabled
UFD session-id                    : 10
UFD Links-to-Monitor status       : Up
Up Delay                          : 20 sec
Down Delay                        : 10 sec
Links-to-Monitor                  : None
Links-to-Disable                  : None
Last Links-to-Monitor Down Time   : None
UFD session-id                    : 20
UFD Links-to-Monitor status       : Up
Up Delay                          : 0 sec
Down Delay                        : 0 sec
Links-to-Monitor                  : None
Links-to-Disable                  : None
Last Links-to-Monitor Down Time   : None
*********************************
Command : show running-config ufd
*********************************
ufd enable
ufd session-id 10
    delay down 10
    delay up 20
    exit
ufd session-id 20
    exit
====================================================
[End] Feature ufd
```

```
==================================================

==================================================
Show Tech commands executed successfully
==================================================
```

Command History

Release	Modification
10.09	Command introduced.

Command Information

Platforms	Command context	Authority
All platforms	Operator (`>`) or Manager (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

`show ufd`

Syntax

```
show ufd [session-id <ID>]
```

Description

Shows information on all UFD sessions or the specified UFD session.

Parameter	Description
`<ID>`	Specifies an existing UFD session ID. Range: 1 to 128.

Example

Showing information on all configured UFD sessions:

```
switch# show ufd
Global UFD Status : Enabled
UFD session-id                     : 1
UFD Links-to-Monitor status        : Up
Up Delay                           : 10 sec
Down Delay                         : 10 sec
Links-to-Monitor                   : 1/1/1,1/1/2
Links-to-Disable                   : 1/1/11,1/1/12
Last Links-to-Monitor Down Time    : 2021-11-03 15:22:05:37
UFD session-id                     : 2
UFD Links-to-Monitor status        : Up
Up Delay                           : 5 sec
Down Delay                         : 5 sec
Links-to-Monitor                   : lag18-lag20
Links-to-Disable                   : 1/1/3-1/1/5
Last Links-to-Monitor Down Time    : 2021-11-01 12:14:42:56
```

Showing information on UFD session 2:

```
switch# show ufd session 2
UFD session-id                     : 2
UFD Links-to-Monitor status        : Up
Up Delay                           : 5 sec
Down Delay                         : 5 sec
Links-to-Monitor                   : lag18-lag20
Links-to-Disable                   : 1/1/3-1/1/5
Last Links-to-Monitor Down Time    : 2021-11-01 12:14:42:56
```

Command History

Release	Modification
10.09	Command introduced.

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

`ufd enable`

Syntax

```
ufd enable
no ufd enable
```

Description

Enables UFD (Uplink Failure Detection). UFD is disabled by default. This command must be issued before the configuration that is set with related UFD commands takes effect.

The no form of this command disables UFD.

Examples

Enabling UFD:

```
switch(config)# ufd enable
switch(config)# ufd session-id 1
switch(config-ufd-1)# links-to-monitor 1/1/1,1/1/2
switch(config-ufd-1)# links-to-disable 1/1/11,1/1/12
switch(config-ufd-1)# delay down 10
switch(config-ufd-1)# delay up 10
switch(config-ufd-1)# exit
switch(config)#
```

Disabling UFD:

```
switch(config)# no ufd enable
```

Command History

Release	Modification
10.09	Command introduced.

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution righ ts for this command.

`ufd session-id`

Syntax

```
ufd session-id <ID>
no ufd session-id <ID>
```

Description

Creates the specified UFD (Uplink Failure Detection) session and then enters its context. If the specified session already exists, this command enters its context.

The no form of this command deletes the specified session configuration.

Parameter	Description
`<ID>`	Specifies the UFD session ID. Range: 1 to 128.

Examples

Creating UFD session 1 and then entering its context:

```
switch(config)# ufd enable
switch(config)# ufd session-id 1
switch(config-ufd-1)# links-to-monitor 1/1/1,1/1/2
switch(config-ufd-1)# links-to-disable 1/1/11,1/1/12
switch(config-ufd-1)# delay down 10
switch(config-ufd-1)# delay up 10
switch(config-ufd-1)# exit
switch(config)#
```

Creating UFD session 2 and then entering its context:

```
switch(config)# ufd session-id 2
switch(config-ufd-2)# links-to-monitor lag18-lag20
switch(config-ufd-2)# links-to-disable 1/1/3-1/1/5
switch(config-ufd-2)# exit
switch(config)#
```

Deleting UFD session 1:

```
switch(config)# no ufd session-id 1
```

Command History

Release	Modification
10.09	Command introduced.

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution righ ts for this command.

Support and Other Resources

Access HPE Aruba Networking support and updates, and view warranty and regulatory information

Subtopics

- **Accessing HPE Aruba Networking Support**
- **Accessing Updates**
- **Warranty Information**
- **Regulatory Information**
- **Documentation Feedback**

Accessing HPE Aruba Networking Support

HPE Aruba Networking Support Services	**https://www.hpe.com/us/en/networking/hpe - aru ba - networking - support - services.html**
AOS - CX Switch Software Documentation Portal	**https://arubanetworking.hpe.com/techdocs/Aruba DocPortal/content/new-portal/aoscx.html**
HPE Aruba Networking Support Portal	**https://networkingsupport.hpe.com/home**

North America telephone	1 - 800 - 943 - 4526 (US & Canada Toll - Free Number) +1 - 650 - 750 - 0350 (Backup—Toll Number)
International telephone	**https://www.hpe.com/psnow/doc/a50011948enw**

Be sure to collect the following information before contacting Support:

- Technical support registration number (if applicable)
- Product name, model or version, and serial number
- Operating system name and version
- Firmware version
- Error messages
- Product-specific reports and logs
- Add-on products or components
- Third-party products or components

Other useful sites

Other websites that can be used to find information:

HPE Aruba Networking Developer Hub	**https://developer.arubanetworks.com/hpe - aruba - networking - aoscx/docs/about**
Airheads social forums and Knowledge Base	**https://community.arubanetworks.com/**
AOS - CX Software Technical Update channel on You Tube.	Videos on new features introduced in this release : **https://www.youtube.com/playlist?list=PLsYGHuNuBZcbWPEjjHuVMqP - Q_UL3CskS**
HPE Aruba Networking Hardware Documentation and Translations Portal	
HPE Aruba Networking software	**https://networkingsupport.hpe.com/downloads https://networkingsupport.hpe.com/downloads**
Software licensing and Feature Packs	**https://licensemanagement.hpe.com/**

End - of - Life information	**https://networkingsupport.hpe.com/end - of - life**

Accessing Updates

You can access updates from the HPE Aruba Networking Support Portal at **https://networkingsupport.hpe.com**.

Some software products provide a mechanism for accessing software updates through the product interface. Review your product documentation to identify the recommended software update method.

To subscribe to eNewsletters and alerts:

https://networkingsupport.hpe./notifications/subscriptions (requires an active HPE Aruba Networking Support Portal account to manage subscriptions). Security notices are viewable without an HPE Aruba Networking Support Portal account.

Warranty Information

To view warranty information for your product, go to **https://www.arubanetworks.com/support-services/product-warranties/**.

Regulatory Information

To view the regulatory information for your product, view the Safety and Compliance Information for Server, Storage, Power, Networking, and Rack Products, available at **https://www.hpe.com/support/Safety-Compliance-EnterpriseProducts**

Additional regulatory information

HPE Aruba Networking is committed to providing our customers with information about the chemical substances in our products as needed to comply with legal requirements, environmental data (company programs, product recycling, energy efficiency), and safety information and compliance data, (RoHS and WEEE). For more information, see **https://www.arubanetworks.com/company/about-us/environmental-citizenship/**.

Documentation Feedback

HPE Aruba Networking is committed to providing documentation that meets your needs. To help us improve the documentation, send any errors, suggestions, or comments to Documentation Feedback (**docsfeedback-switching@hpe.com**). When submitting your feedback, include the document title, part number, edition, and publication date located on the front cover of the document. For online help content, include the product name, product version, help edition, and publication date located on the legal notices page.